

Review

Sustainable management of organic agricultural wastes: contributions in nutrients availability, pollution mitigation and crop production

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Abstract

Recent climate variability, limited fertile land availability and soil degradation are major constraints to achieve food security for an ever-increasing human population. The adoption of intensive agriculture practices and highly productive agrosystems coupled with intensive use of agrochemicals has caused significant increases in agriculture production worldwide. This exhaustive production system has caused significant increases in generation and accumulation of large quantities organic wastes, which cause environmental pollution, deteriorate soil health and increase public health hazards. Conventional methods of organic waste management such as in situ burning of organic wastes, landfilling and chemical degradation are labour intensive, expensive and highly energy-consuming, and negatively impact the environment. Therefore, sustainable, eco-friendly and socially acceptable agri-technologies have been developed for value-added management of organic wastes and to obviate pollution problem. These biological technologies such as composting, anaerobic digestion, vermicomposting, production of biochar, organic phytostimulants, and bioremediation of pollutants have opened new vista for organic waste management. The addition of processed organic amendments (i.e., compost, vermicompost, biogas slurry and biochar) to the soil increases soil organic matter and nutrients availability, stimulates soil microbial community, contributes towards biocontrol of pathogens and also causes detoxification of pesticides. Furthermore, soil amendment with processed organic waste material, singly or with beneficial microbes, improved soil health, promoted growth of plants and increased the crop yields with less dependency on chemical fertilizers. In this article, the current technologies used for management of accumulated organic wastes are discussed for improving sustainable crop production, while maintaining environmental sustainability.

Highlights

- Intensive agriculture practices produce significant quantities of organic agricultural wastes.
- Composting, vermicomposting and biochar production are sustainable technologies for organic waste management.
- Fertilization with organic amendments increases soil organic matter, nutrients availability, and mitigates abiotic as well as biotic stresses.
- The combining of compost or biochar with beneficial microbes promotes plants growth and increases crop yields.
- More *in situ* field experiments are needed to manage accumulated organic wastes for improving sustainable crop production.

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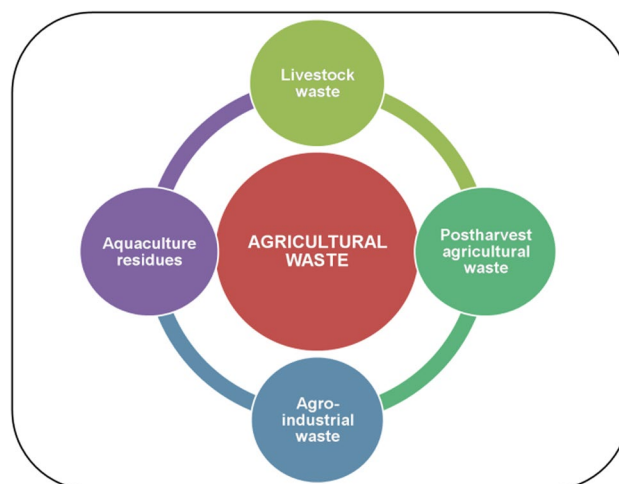
1 Introduction

The food production targets must be increased by 60–100% for feeding an exponentially increasing 9.7 billion global population by 2050 [1, 2]. Therefore, intensive agriculture productive agrosystems coupled with novel agri-technologies are adopted worldwide in achieving food security goals. However, the limited fertile land and accessible natural resources, soil degradation, depletion of soil organic matter (SOM), nutrients imbalance in soil and drastic climate changes are the major constraints for crop production and food security worldwide [3, 4]. In addition, injudicious use of agrochemicals (mainly chemical fertilizers and pesticides) and plant growth enhancers in these intensive agriculture practices has resulted in environmental pollution and community health threats [5, 6]. Therefore, there is an exigent need for climate-adapted sustainable agricultural practices for sustained food production from cultivable land. With significant increases in crop yields by using novel agri-technologies recently, large quantities of agricultural wastes or organic wastes (OWs) are also produced globally [7, 8].

Organic wastes contain materials, which have originated from living organisms and are biodegradable. Organic wastes comprises of agricultural waste, municipal solid waste, industrial solid waste and wastewaters [9, 10]. On their source of origin, agricultural wastes are categorised into four major types, which includes livestock waste, agricultural waste, agro-industrial residues and aquaculture wastes (Fig. 1). Around 120 million tons of waste is produced annually from livestock sector, which generally refers to cattle, poultry and pig farm waste including urine, dung, wastewater, waste feed, feather, hoof, horn, bedding material etc., and waste from slaughter houses [11]. In addition, about 200 billion tons of crop plant residues and plant biomass is formed annually in the world, nearly 90% of which is lignocellulosic [10, 12]. Post-harvest agricultural waste is considered as primary agricultural waste and it includes leaf litter, straw, husk, weeds, seed pods and plant stalks. Cellulose that accounts for 40–50% of the plant dry weight is the main constituent of lignocellulosic waste, and is rich in organic carbon and other plant nutrients. Agro-industrial residues are considered as secondary agricultural wastes as these are the industrial by-products of primary agricultural residues and waste products from food processing units. They generally include bagasse, molasses, pulp and peels of different fruits and vegetables, oil seed cakes etc. In recent years, aquaculture waste is also considered as a type of agricultural wastes containing aquaculture residues such as uneaten feed and their fecal waste [8, 10].

Accumulation of vast amount of agricultural wastes in soil along with the build-up of toxic compounds, heavy metals (HMs), pesticides, salts, and pathogens causes soil pollution that adversely affect health of plants, animals and humans, and also cause major economic losses [6, 13, 14]. Contaminated/polluted soil is not ideal for agricultural, industrial or other recreational activities as it reduces soil fertility and negatively impact crop productivity [15, 16]. Heavy metals like lead, arsenic, mercury, nickel, zinc and copper are also being introduced in the soil through different agricultural activities such as long-term application of various agrochemicals and urban sewage sludge in crop fields, waste incineration

Fig. 1 Different categories of agricultural waste based on their source of origin



and release of untreated waste in the farm and environment [17, 18]. All these HM pollutants are potential health risks to microbes, plants, animals and humans, because of their transfer from soil to plants [19–21]. Farmers generally leave agricultural waste in the fields or burn them. Burning of agricultural residues releases 141.15 Mt of CO₂, 8.77 Mt of CO, 0.12 Mt of NH₃ and 0.23 Mt of NO [22] and it contributes to global warming. This also causes soil contamination and loss of soil organic matter (SOM) content to the tune of 80–90% N, 25% P and 20% K [12, 23]. Thus, depletion of SOM, HM pollutants, salinization and nutrient imbalance in soil affect soil microbial population and their beneficial activities, nutrient cycling and physical characteristics of soil [4, 24] and many agro-practices have been proposed to improve the level of SOM for ensuring sustainable fertility and crop productivity of degraded soil.

Conventional methods of solid waste management include thermal processes (incineration, gasification or uncontrolled pyrolysis), chemical degradation and landfilling. These high cost and energy consuming processes negatively impact the environment by discharging a variety of harmful chemicals and greenhouse gases. These problems have necessitated development of sustainable, economical and socially acceptable organic waste management techniques for management of solid organic waste, which may help in increasing the economic value of the discarded organic wastes [25]. Therefore, biological conversion of agricultural waste has emerged as a sustainable and economic option, and addition of processed OWs to soil enhances the nutrient content and fertility of soil [26–28].

Several technologies have been developed for biodegradation of OWs such as composting, vermi-composting, anaerobic digestion, production of biochar and organic phytochemicals etc. [29–31]. These biological technologies utilize different indigenous and exogenous organisms such as bacteria, fungi, algae, protozoa, nematodes and arthropods to decompose and process agricultural waste [32–34]. For instance, olive mill wastewater sludge generated by the olive oil industry was converted into highly beneficial organo-mineral fertilizers after processing [35, 36] and its addition improved soil properties, including water availability, and the nutrient concentrations, such as P, K, and magnesium (Mg). Furthermore, addition of organo-mineral fertilizer (OMF) formulations on *Zea mays* positively affected the plant's respiration, photosynthetic activity, and nutrient assimilations [37]. Organic wastes can also be used for energy recovery (such as heat, liquid fuels, biohydrogen, or electricity), as well as for the production of certain chemicals, such as volatile organic acids and alcohols [38–40]. Microbial decomposition of OWs under aerobic and anaerobic conditions is the most efficient method of decomposition, and single or co-application of organic amendments such as compost and biochar has been reported to stimulate microbial community and improved the fertility of deteriorated soil with low organic matter content [41–43]. Also, many studies have focused on the development of agricultural waste based biofertilizers to replenish macro- and micro-nutrients in polluted soils and to minimize the use of agrochemicals in farmlands [33, 44]. Another sustainable approach to mitigate soil pollution is bioremediation, which is used for removal or stabilization of HMs and decontamination of pesticides in polluted soil [21, 45, 46]. Bouhia et al. [32] identified potent native fungal species, FUN 06 (*Galactomyces geotrichum*), within olive mill wastewater sludge (OMWS) suitable for bioremediation and its transformation into a high-value organic fertilizer using a mixture of OMWS and rock phosphate (RP).

While there have been numerous research papers and reviews that discuss the approaches used for organic waste management [26, 30–32]. However, there has been little comprehensive discussion about the potential contribution of processed OWs and associated beneficial microbes in different biological processes, pollution mitigation and crop production (although see, 25, 37, 47). In the present manuscript, a broad overview has been provided about the main strategies employed for management and processing of different kinds of OWs in improving nutrition biavailability, functioning of various biological processes, bioremediation of soil pollutants and for increasing crop productivity in a sustainable way. A more in-depth discussion about novel approaches and strategies has been provided about various mechanisms of OWs processing and microbial communities for increasing their efficacy to use processed OWs as organic fertilizers for environmental sustainability.

2 Need for soil fertilization with exogenous organic matter

Earth's soil resources provide nutrients for growth of plants and soil-inhabiting microorganisms. Due to recent intensive agricultural practices and soil erosion, agricultural soils are degrading faster with reduction in organic matter than it is naturally replenished. In addition, the uncontrolled use of agrochemicals and mounting apprehensions over contamination of the environment has increased the need for safe and nutritious food. Therefore, future of humanity over the next century is severely threatened by unsustainable soil management practices [48]. Thus, organic farming has gained significant global attention recently to maximize the agronomic efficiency and crop productivity [49]. Organic farming practice employs processed organic waste products for achieving optimal agricultural productivity with conservation of

agroecosystems and proper functioning of soil ecosystems, while minimizing adverse impacts on both the environment and human communities on soil health and fertility [50, 51].

Soil-inhabiting microorganisms play important role in nutrients' solubilization and bioavailability for plants, formation of soil organic carbon (SOC), its preservation and loss [52, 53]. Soil organic carbon in living soil is a major engine of terrestrial biogeochemistry, which regulates the turnover of terrestrial C pool and soil organic matter, act as a source of plant nutrients for ecosystem productivity, provide diverse energy sources for soil microorganisms to express beneficial activities, act as fundamental control on soil structure, and plays a critical role in terrestrial ecosystem functioning [54–56]. Unfortunately, the level of SOC is quite low in most upland soils and it ranges from 0.5% to 3.0% organic C in the topsoil. Therefore, soil fertilization with exogenous organic matter (EOM), including composts, vermicomposts or biochar of organic waste origin prepared from agriculture, forestry or industry, is considered essential in improving the fertility and organic matter content of soil for sustainable development and circular economy around the world [57].

Soil organic matter significantly improves the soil's capacity to store and supply essential nutrients (such as N, P, K, Ca and Mg), and to retain toxic elements. It allows the soil to cope with changes in soil acidity, and helps soil minerals to decompose faster. Ayangbenro et al. [58] reported that understanding the continuing transformations of organic amendments and the distribution of different factors (carbon, nutrients, and microbial activities) across and along roots is crucial in the soil and rhizosphere for promoting plant growth and health. Liu et al. [59] showed that organic manure fertilization in wheat greatly decreased the network complexity ratio of the rhizosphere to bulk soil for bacteria and fungi. The proportion of bacteria in the rhizosphere derived from the bulk soil was much higher in comparison to fungi. In another study, addition of exogenous straw as green management practice in maize rhizosphere accelerated nutrient cycling by triggering microbial growth and biomass, and synthesizing a large number of enzymes including β -1,4-glucosidase (BG) and β -N-acetylglucosaminidase (NAG) hydrolases [60]. In contrast, the addition of biochar increased the relative abundance of Actinobacteria, enhanced the root biomass by 31%, increased mobilization of nutrients in the rhizosphere, and expression of BG and NAG was accelerated by 26 and 47%, respectively by expanding the extent of rhizosphere hotspots from a larger soil volume.

Earlier studies showed that addition of SOM regulated soil phosphorus (P) dynamics and producing phytoavailable P in soils for effective management of soil health and enhancement of soil fertility, especially P-use efficiency [61]. Massaccesi et al. [62] observed stimulation of the microbial community and increases in the enzymatic activities in compost (CMP) amended soils, indicating that the organic matter provided by the compost is more readily mineralized than that added with the biochar. Moreover, the biochar (BIO) amended soils showed the lowest $q\text{CO}_2$ and ecoenzymatic C:N ratio, and the highest $q\text{MIC}$ (1.4%), indicating a C immobilization process into the microbial biomass, which need increases in N-cycling enzyme activity. In the short term, a boost of soil C and N cycling was evident with CMP and BIO amendments, respectively. In another experiment, Yang et al. [63] demonstrated that incorporation of organic amendment (Grass peat) improved soil nutrients, organic C sequestration, water use efficiency (WUE) and yield of maize (*Zea mays* L.) under dryland agriculture in China. In a recent study, two green manure treatments [*Vicia villosa* (hairy vetch) and *Brassica juncea* (L.)] were applied with and without *Trichoderma* spp. strain RW309, and control (soil only) [64]. Plant growth was significantly higher with green manure combined with RW309 inoculation than with green manure alone. The enzyme activities and fungal biomass were also markedly higher in the co-application treatments than in the other treatments. Moreover, co-incorporation of green manure and RW309 enhanced specific fungal genera *Penicillium* spp., which further stimulated mineralization of nutrients, enhanced plant growth, and improved soil health for sustainable agricultural practices.

3 Effect of agricultural wastes and pollutants on physiology and functioning of crop plants

Soil is the fundamental requirement for sustaining life on earth [53] and it is constantly changing and developing with time in response to environmental factors. Some of these changes are short-termed and reversible, while some become permanent characteristics of the soil. Increased agricultural production required for exponentially growing population has led to large accumulation of agricultural wastes and their by-products, and causing soil pollution and soil degradation [65, 66]. In addition, litter quality has also been demonstrated to control the contribution of microbial carbon to main microbial groups and SOC during its decomposition [67]. Moreover, the burden of climate change mediated environmental stresses can be easily seen on the environment especially on soil [68, 69]. Furthermore, the buildup of

toxic pollutants in soil negatively affects the physico-chemical and biological features of soil and adversely affects the growth and development of plants, resulting in lowering the crop yield [14, 65, 70, 71].

Different pollutants including organic waste, industrial waste (such as molasses, bagasse, and processed food waste), agrochemicals (such as pesticides, insecticides, herbicides and chemical fertilizers) and HMs (mainly lead mercury, cyanides, copper, zinc, cadmium etc.) contribute towards soil pollution [17, 65]. Many of the pesticides and herbicides are still used for agricultural purposes in developing countries to control the pests for increasing the crop yield, but their injudicious use alters the biodiversity of the soil [71]. These pesticides include highly persistent and toxic chemicals such as aldrin, chlordane, DDT, dieldrin, endrin, heptachlor, mirex, polychlorinated biphenyls, polychlorinated dibenzo-p-dioxins, polychlorinated dibenzofurans and toxaphen [72, 73]. Industrial products such as detergents, petroleum hydrocarbons, plastics, organic solvents, pesticides, and dyes, are the vital sources of organic pollutants and are the additional sources of soil pollution. Some other organic pollutants in wastewater include phenols, oil, surfactants, pesticides, herbicides, excreta of animals etc. [32, 74]. These organic pollutants are very slowly degraded by microflora inhabiting soil and water; hence, they keep on accumulating in the soil. As a result, they negatively affect the plant growth by inhibiting their overall growth, while reducing the yield and size of the grain and fruit.

Presence of HMs in soil at low concentrations have little effect the plant growth, but after a threshold limit, the plants will be poisoned by these HMs leading to their death [18, 75, 76]. In a study by Sandalio et al. [77] observed reduced plant growth in soils contaminated with high levels of cadmium (Cd) due to decreased nutrient uptake from contaminated soil, inhibition of various enzyme activities and initiation of oxidative stress in plants. Bini et al. [78] reported negative effects on root and shoot fresh weight by application of wastewater, which was polluted with HMs and other organic pollutants. The polluted wastewater caused reduction in water and nutrient uptake by roots leading to decreased relative water content and subsequently damaged the roots leading to retardation effects on plant growth.

Generally, plants absorb the nutrients and water along with pollutants and/or their degradation products from soil, and from plants, these pollutants enter food chains; hence affecting health of animals and humans [6]. On the other hand, fertile and healthy soils provide sufficient nutrients for crop plants resulting in increased nutrient-rich crop yields, whereas contaminated soil leads to poor crop yield with decreased nutritional value. Ahmed et al. [79] reported that irrigation of turnip and lettuce plants with industrial waste water negatively influenced all the measured yield components like number of pods, fresh weight per plant, seed number and weight per plant, and seed index. In some cases, livestock and human excreta is also used directly as organic manure to enhance crop production but its addition may result in soil pollution and its contamination with various pathogens, which may be present in the excreta. The number of beneficial bacteria, fungi and earthworms present in soil that bind it together start to decline, changing the soil structure and creating an additional problem of soil erosion.

4 Sustainable strategies for management of organic waste and pollution mitigation in soil

Due to mounting apprehensions over contamination of the environment resulting from the uncontrolled use of agrochemicals and increasing need for safe and nutritious food, organic farming has gained significant global attention recently [49]. Organic farming practice employs processed OW products (including composting, vermi-compost, biogas slurry and biochar) for achieving optimal agricultural productivity with conservation of agroecosystems and proper functioning of soil ecosystems, while minimizing adverse impacts on both the environment and human communities [50]. Therefore, OW management has emerged as an issue of concern for researchers and environmentalists involved in ecological and sustainable agriculture all over the world [80, 81]. In comparison to conventional approach using unprocessed or untreated OW, a good organic waste utilization technology rapidly converts the organic waste residues into readily available, nutrient rich material without producing any harmful and unsuitable end products [32, 82, 83]. This necessitates the use of better technology and incentives, and a better approach to organic waste management. According to USDA agricultural waste management field handbook, an agricultural waste management system is a well-planned system designed to regulate and use agricultural by-products in such a way that maintains or upgrades the quality of the environment (air, water, soil, plant, and animal resources) [84, 85]. Such systems are designed and planned to manage all types of agricultural waste for utilization throughout the year.

An organic waste management system is a function of six fundamental steps, including waste production, collection, storage, treatment, utilization and transfer as well as supervision of waste material throughout the process [86]. It starts with the production and accumulation of different waste and residues followed by their collection from original production source or from dumping site. Next, the waste is temporarily stored in different storage facilities until its treatment.

Different activities like analyzing the features of the waste, figuring out the preferred features of the waste after treatment, selecting the kind of treatment facility, as well as its projected size, location, and installation costs, and calculating the total cost of the treatment process must be considered for waste treatment. After appropriate treatment and processing of waste material, OWs are utilized as organic fertilizers for enhancing crop productivity along with mitigation of soil pollution. Another important step is the transfer of waste material throughout the process i.e., from source to collection site, to storage facility, to treatment plants and at last to the site of implication [85]. It also involves the supervision and after-care of disposal sites so that the waste and/or pathogens from the waste do not migrate into the population.

Till now, most of the research to utilize OWs has focused on conversion of these wastes for fertilizers, bio-energy and biogas production [25, 87–89]. For instance, Babu et al. [39] revealed that biomass from agricultural waste may be used to create second-generation biofuels and may serve as a fodder for animals and raw material industrial products. In organic production systems, composted agricultural waste can replace energy-intensive chemical fertilizers [31, 90]. Thus, development of environmentally robust agri-waste residue/OW management technologies has the potential to mitigate soil pollution in situ and ex situ, and contribute towards establishing a green economy in the area of agricultural production. Composting, vermi-composting and biochar production are some of the predominant methods used in recycling of organic wastes, which are subsequently used in sustainable agriculture for improving crop production [91, 92]. Various biological processes used in OW management are briefly discussed in this section.

4.1 Production of compost from organic wastes

Composting process provides the most efficient way of handling agricultural OWs and it has emerged as a basic and valuable technique for preparing a nutrient rich fertilizer from OW materials [26, 37, 93]. Thus, OWs may be effectively recycled in the system itself as organic manure that acts as a soil conditioner in agricultural soils improving soil's fertility, structure, water holding capacity and buffering capacity [94]. Therefore, the process of composting is advantageous for disposing OW materials from an ecological and financial standpoint. Composting involves the aerobic/anaerobic and mesophilic/thermophilic breakdown of putrescible in organic waste by microorganisms and certain invertebrates [9, 95]. This process has also proven to enhance pesticide degradation in soil as well as in compost heaps [96, 97]. The final product of composting is a stable organic material, which is particularly relevant to agricultural production and management of forest lands. Application of compost in soil has also helped in reducing loss of N from soil as compared to application of fresh manure. Recently, Hang et al. [93] showed that fertilizers produced through composting process contain higher organic matter, averaging 45.42%, compared to commercial organic fertilizers and bio-organic fertilizers. It was predicted that organic matter content could be increased to 49.3% by optimization of equipment and composting processes in Chinese organic fertilizer companies, which will potentially reduce annual carbon emissions by an estimated 3.07 to 6.97 billion kg of CO₂ equivalent.

Based on the raw material used for the process, compost can be classified as green compost and brown compost. Green compost is prepared from plant byproducts such as crop residues, leaf and litter waste and yard wastes. Brown compost is prepared mainly from wet wastes such as animal manure, kitchen wastes and municipal organic waste [98]. Compost can also be classified as aerobic composting and anaerobic composting depending on the process conditions. For instance, Foughal et al. [89] inoculated a mixture of agri-food waste including fish, vegetable, and poultry wastes with *Aspergillus niger* for accelerating waste biotransformation in a 20-day process. A combination of all three waste types with a tiny excess of poultry waste was found as the best, effective and secure mixture for biofertilization. Anaerobic digestion product had a higher potential for biofertilization than aerobic biotransformation product, according to fertilization tests on tomato crops.

4.1.1 Aerobic composting

Composting done under aerobic conditions is the most popular and efficient technique of converting OWs into compost (Fig. 2), which produces mature compost within a short time period [99]. During the process, the raw waste material is biologically converted to a stable, pathogen-free and odour less product ready to apply in agriculture and forestry [100, 101]. The product can be obtained within around 4 weeks with regular turning of compost pile [91]. However, if the compost is not turned regularly, it may affect the quality and duration of compost. Sufficient amount of oxygen and aerated conditions are required during the aerobic composting process. A compost pile containing organic waste, sufficient amount of moisture with balanced carbon and nitrogen ratio (C:N) is set up in pits or in an open area. Mesophilic microorganisms (ideal temperature ranges from 20 to 45 °C) start to multiply in the

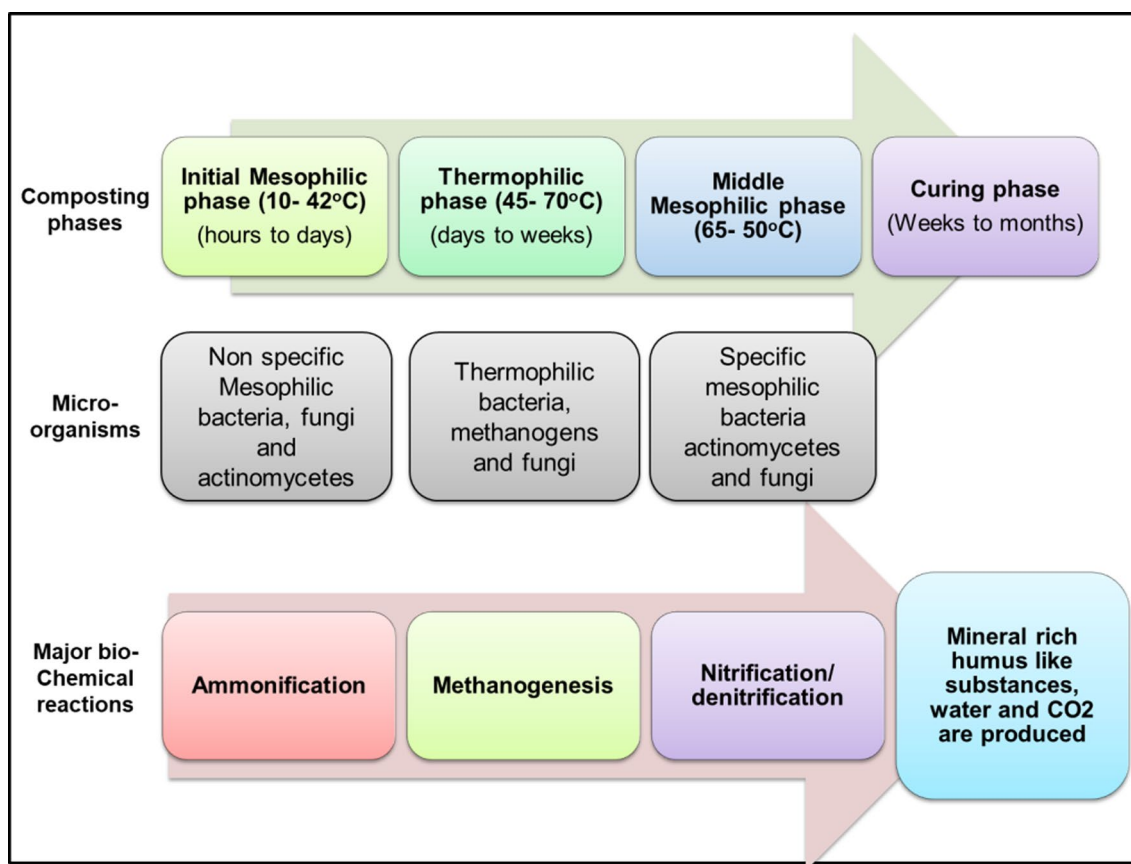


Fig. 2 Graphical representation of the composting process along with the microorganisms involved and major biochemical reactions

pile metabolizing complex polysaccharides, while producing heat [26, 102]. With increase in the temperature of the compost pile, it becomes ideal for growth and functioning of thermophilic microorganisms (ideal temperature 50 to 70 °C) [102]. Further increases in the pile temperature (to > 62 °C), destroys the pathogens and weed seeds present in the compost, hence improving compost quality [103]. After decomposition of major OWs, the pile temperature reduces gradually, allowing the mesophiles to take over the process once again [104]. The last step in this process is curing. Curing phase occurs when the pile stops reheating even after turning the pile. This step indicates completion of the composting process. The final product is a dark-brown or black with reduced particle size and it lacks active microorganisms. Mineralization and humification are the main steps of degrading organic matter through composting [103]. The time duration of compost formation varies with different technologies. Most popular composting methods include Indian Bangalore composting, windrow composting, static composting, sheet composting, Indian Indore composting and Berkley rapid composting [9].

The efficiency of different technologies for composting OWs depends on the raw materials used, microbial inoculum, site of composting, climatic conditions, moisture content, aeration conditions and dimensions of the heap [102, 103]. The current knowledge on improving the efficacy of composting due to inoculation of efficient microorganisms in a compost pile containing OWs, and the avenues for future research, leading to a holistic view and understanding of nutrient-rich compost are discussed in this section. Strong microbial activity during composting improves the quality of the nutrients in the compost by the activity of nitrifiers, ammonia oxidizers, sulphur oxidizers, and nitrogen-fixing microorganisms [104, 105]. For instance, inoculation of sugarcane trash and rice straw with cellulolytic microorganisms such as *Aspergillus awamori*, *Trichoderma viride*, *Phanerochaete chrysosporium*, *Trichorus spiralis* and *Paecilomyces fusisporus* enhanced the composting process and produced composts with lower C:N ratio. In addition, inoculation with N₂-fixing *Azotobacter*, phosphate-solubilizing fungus *Aspergillus awamori* and cellulolytic fungi along with rock phosphate (RP) helped to produce composts from plant residues in shorter period and with highest N and P content. Bouhia et al. [106] showed that *Pichia* was the most represented genus with almost 53% of the total identified fungal population in raw olive mill wastewater sludge composting. Moreover, the bacteria that dominated were *Zymobacter*

palmae (20%) and *Pseudomonas* sp. (19%). Correlation analysis revealed that *Candida* sp., *Galactomyces* sp. and *Pichia manshurica* showed the greatest influence on the composting process.

Yu et al. [107] reported that during the composting of rice straw, CO₂ emissions were decreased by the addition of nicotinamide adenine dinucleotide (NADH) and adenosine triphosphate (ATP). The structural equation model revealed that NADH may enhance cellulose breakdown into organic compounds and soluble sugar, which are turned into humus, while ATP may support lignin conversion into humus via the sugar-amine condensation pathway and lignin-protein pathway. Fu et al. [108] reported that addition of maleic anhydride (MAH) improved the activities of laccase and tyrosinase, and thus enhanced the catalytic conversion of humus precursors in composting. The analysis of bacterial community showed that MAH optimized the community structure of humification functional enzymes producing bacteria, with the most obvious increase of Firmicutes. Microbial inoculants i.e., rice straw plus water plus Pusa decomposer (RSWF), and rice straw plus water plus TNAU (Tamil Nadu Agricultural University, India) biomineralizer (RSWB) accelerated the degradation of agricultural biomass [34]. When rice straw was subjected to microbial treatment, it decayed rapidly as compared to the control treatments i.e. rice straw (RS) alone and rice straw combined with water (RSW). The C:N ratios for RSWF and RSWB were 20.83 and 33.85, respectively, less than those for RSW (66.11) and RS (78.28), suggesting that microbial consortia could be added to the soil for faster decomposing of rice straw leading to improvement in soil fertility.

In similar studies, Li et al. [109] assessed the breakdown of a compost mixture made of sawdust, poultry manure, biochar and vegetable waste by using *Clonostachys rosea*, *Bacillus amylolyticus* and *Rhodospirillum photometricum*. After inoculation of the raw materials with microbial consortium, the temperature of the compost increased, the thermophilic period lengthened, and the rate of cellulose breakdown was improved. The variety and abundance of decomposing microbial colonies was also changed with the inoculation of microbes. The consortium of *C. rosea* and *B. amylolyticus* proved to be better as compared to other consortia, suggesting that by improving the physical and chemical environments of organic wastes and encouraging the variety and multiplication of helpful bacteria engaged in the degradation of cellulose, microbial agents can improve the composting process. The inoculation of lignocellulose degrading strains b4 (*Bacillus subtilis*) and z1 (*Aspergillus fumigatus*) caused up-regulation of lignocellulose degrading enzyme genes during rice straw composting [110]. Modular microorganisms, such as Actinobacteria, Proteobacteria, Ascomycetes and Basidiomycetes, were identified as driving factors that affected lignocellulose degrading enzyme genes abundance. On the other hand, pH, organic matter and soluble sugar contents affected lignocellulose degradation and growth of modular microorganisms during rice straw composting.

Dasgupta et al. [111] assessed the decomposition dynamics of ¹³C and ¹⁵N-labeled wheat stubble using four treatments i.e., soil-only, soil + inorganic nutrients, soil + wheat-straw, and soil + wheat-straw + inorganic nutrients. The addition of straw markedly increased microbial biomass C and fungal: bacterial ratio. Treatments significantly differed in structuring the microbial communities, with a stronger effect on bacteria than fungi. Straw and nutrient amendments also enhanced the complexity of the core microbiome. Members of *Devosiaceae* were significantly enriched upon inorganic nutrient addition and were also positively associated with microbial biomass C. During leaf litter decomposition of *Pinus massoniana* in a subtropical forest, Keyhani et al. [112] showed that mineral fertilization (fertilization with N and P or N, P, and K) improved both shoot as well as belowground root components, and significantly affect bacterial diversity than fungal diversity as compared to control. Proteobacteria, Firmicutes, Bacteroidetes, Acidobacteria, and Cyanobacteria responded positively to fertilizers, while only Ascomycota and Basidiomycota were influenced among fungal phyla. Redundancy analysis indicated shifts in community structure linked to soil chemical properties. In another recent study, effects of microbial agents were evaluated on N conversion during cotton straw composting [113]. Microbial inoculation enhanced the abundance of *nifH*, *glnA*, and *amoA* genes and reduced the major denitrification genes (*nirK*, *narG*, and *nirS*). Network analysis model showed distinct changing patterns of bacterial communities after inoculation, and identified keystone microorganisms in maintaining network complexity and system function to promote N preservation. Microbial inoculation also increased the germination index and total nitrogen by 24–29% and 7–10 g kg⁻¹, respectively.

4.1.2 Anaerobic composting

Anaerobic composting is carried out in the absence of oxygen in an enclosed system, commonly known as digesters, by utilizing anaerobic microorganisms. During this process, a mixture of different gases (primarily methane and carbon dioxide, with traces of hydrogen sulfide and hydrogen), collectively known as biogas, is produced [94]. The methane content determines biogas' calorific value, while CO₂ acts as a contaminant. The spent slurry or compost can be used as manure in agriculture and forestry [114]. During anaerobic composting, superior compost and liquid fertilizer per unit volume of raw material is produced as there is minimal nutrient loss as compared to aerobic digestion. However,

according to FAO [115], this process takes longer time as compared to aerobic composting. The final product is characterized by a non-pleasant smell resulting from production of higher levels of hydrogen, ammonium, sulfur and biogas. Another major disadvantage of this process is that it leaves many pathogens and weed seeds undestroyed, since higher temperatures are not achieved during the anaerobic process. However, due to very hostile conditions, the pathogens and weeds die gradually. Also, this is a costly and labor-intensive process, especially when applied on a large scale [116].

Anaerobic digestion (AD) of organic waste and biomass involves a series of degradation processes carried out by microorganisms in the absence of oxygen, resulting in production of biogas and nutrients rich slurry or digestate [114]. The critical physico-chemical parameters that need to be assessed for optimal AD performance include contents of total solids (TS), volatile solids (VS), pH, carbon-to-nitrogen ratio (C/N), and biochemical composition. The biogas slurry contains water (93%) and dry matter comprising of 4.5% organic matter and 2.5% inorganic matter. In addition, significant amounts of both macronutrients (N, P, K) and micronutrients (Zn, Mn, B) are found in biogas slurry. Due to availability of nutrients in the slurry/digestate, it is applied as a biofertilizer in agriculture and enhances plant growth. Therefore, many small-scale anaerobic digesters were installed in several Asian countries leading to significant increases in production of biogas and biogas slurry [117]. Depending upon the optimal degradation of organic waste by microorganisms, the digested slurry also contains readily available plant nutrients such as amino acids, minerals, and bioactive compounds including vitamins, humic acids, and hormones [118]. Therefore, application of biogas slurry under field conditions is an efficient strategy for recycling of nutrients in agriculture farms, which also reduces the use of chemical fertilizers and environmental problems globally.

Deng et al. [119] used an amino acid-derived ionic liquid, glycine hydrochloride, to enhance methane production from anaerobic digestion of rice straw. The addition of glycine hydrochloride increased bacterial diversity and increased the abundance of methanogens and archaea. The peak activities of cellulase, xylanase, and lignin peroxidase enzymes were significantly higher than those of the control (without additive). The cumulative methane production was 115.56 mL g^{-1} with glycine hydrochloride additive, which was 73% higher than that of the control after 40 days of digestion. Czekala et al. [120] reviewed the effect of reutilizing anaerobic digestate solid fraction on the composting process. The solid digestate functioned as an important substrate for the manufacture of compost, both as a monosubstrate and as a desirable supplement for additional raw materials to supplement organic matter to the soil. Thus, anaerobic digestate solid fraction was established as innovative process-improving additive for improving composting process, and may be applied in land reclamation. Moreover, its implementation in modern bioeconomy will provide effective waste management.

Machine learning (ML) technique was used to predict specific methane yields (SMY) with a dataset of 14 features from lignocellulosic biomass (LB) characteristics and operating conditions of completely mixed reactors under continuous feeding mode [121]. Biomass compositions greatly influenced SMYs from LB, and cellulose prevailed over lignin and biomass ratio as the most important feature. Impact of LB to manure ratio was assessed to optimize biogas production with the random forest (RF) model. Under typical organic loading rates (OLR), optimum LB to manure ratio of 1:1 was identified. Experimental results confirmed successful applications of ML for anaerobic digestion modelling and influential factors were revealed by the RF model that provided the highest SMY of 79.2% of the predicted value. In another study, hydrochar co-compost (HCO) was formed by anaerobic breakdown of municipal solid waste (MSW) combined with hydrothermal carbonization (HTC) [122]. HCO may subsequently be used to recover biogas and compost. Interestingly, hydrochar co-compost cytotoxicity was found similar to standard compost cytotoxicity. Wang et al. [123] conducted their research on the metabolic features of bacterial and fungal groups at the time of composting as well as the use of biochar and biogas leftovers in composting. It was established that about 50% the functional genes involved in carbon fixation, organic matter decomposition and methane oxidation were found of *acsA* (gene encoding acetyl-CoA synthetase) abundance. On the other hand, abundance of denitrification gene *nirK* (encoding nitrite reductase enzyme; is involved in denitrification converting nitrite to nitric oxide) was dramatically reduced by the presence of biogas residue in the compost. Biogas residues were also found to enhance population of bacterial as well as fungal communities. The redundancy analysis showed that pile temperature, pH and EC were the key physico-chemical parameters influencing the expression of microbial community *nirK* gene.

4.1.3 Production of humic substances

Humic substances (HSs) occupy 80% of organic matter in soil and significantly influence soil physico-chemical properties, and have been widely applied for soil remediation agents, and adsorbents [124]. The biochemical reactions involved in aerobic composting degrades organic matter and the products aggregates to produce 'humic like substances' [91, 125]. HSs are a core product of pre-controlled degradation and chemical oxidation of OWs, and a crucial indicator of

the maturation of compost that can be used in soil amendments. These HSs can be obtained from naturally humified organic matter, from composts or vermi-composts or from mineral deposits [126]. Humic substances as well as their compounds in soil (humins, humic acid and fulvic acid) are a result of interaction between organic matter, microbes and plant roots [127, 128]. They can also be formed ex situ by controlled breakdown and oxidation of OWs resulting in humic like substances. Since the HS extraction rate is very low by microbial degradation in nature, artificial humification processes such as aerobic composting and hydrothermal treatment have attracted a great deal of attention as the most important strategies in HS production [124].

The formation of HSs is complex process and is affected by the characteristics of the raw materials involved, the presence of compost additives, microbial activity, temperature, pH, the C/N ratio, moisture content, oxygen content and particle size, all of which can interact with each other. Humic substances can be divided into humic acids (HAs), fulvic acids (FAs), and humins based on their characteristics in alkaline and acidic solutions. However, HSs represent the same functional groups and structural arrangements, which help to predict their structures. Functional groups represented by phenol and carboxylic acid groups of HAs and FAs can provide various agronomic functions, such as plant growth enhancement, water and nutrient retention, and disease suppression capacity. Thus, HSs can act as a soil amendment, fertilizer, and plant growth regulator.

The effect of HSs varies with soil environment, the nutrient accepting plant, the dosage and method of application. Humic substances strongly affect solubility of several plant nutrients such as P, Fe, and Cu, thereby increasing their plant availability [129]. In addition, SOC, mainly HSs, can detoxify monomeric Al in acid soils, can strongly bind toxic heavy metals, making them unavailable to the plant roots, and may strongly bind a vast variety of harmful organic pollutants [124, 130]. The inclusion of farmyard manure (FYM) and its rotted or composted forms were found superior compared to the separate application of straw and slurry to soil. Soils amended with polyanionic HSs shows enhanced cation exchange capacity leading to increased nutrient uptake. Humic substances also stimulate H⁺-ATPase activity for uptake of nitrate and other nutrients. They also help in stress mitigation by phenyl-propanoid metabolism leading to the production of phenolic compounds [91, 124]. These functions of HSs enhance the reuse potential of organic waste compost products; however, this requires scientific control of various composting parameters and appropriate application of final products.

4.2 Decomposition of organic waste material into vermicompost

Vermicomposting is a type of aerobic composting based on the decaying of organic biomass with the help of earthworms and some associated microorganisms. This is a mesophilic process, where the end product i.e., vermicompost is a stable, humus-like material [131, 132]. The species that are used for vermicomposting are usually epigeic [133], and they live in areas containing high amount of organic matter feeding on leaf litter, decaying plant roots or dung. Species of earthworms used in this process remain on the soil surface and do not form permanent burrows. In this process, different earthworm species most commonly used includes *Eudrilus eugeniae*, *Perionyx excavatus*, *Eisenia andrei*, *E. fetida* [132, 134], and *Pontoscolex corethrurus* [135].

With time, many variations in vermicomposting methodology have been developed. It may vary from composting in windrows to completely automated, continuous-flow reactor arrangements [136]. The process of vermicomposting occurs in two phases: (i) the active decomposition phase where the earthworms ingest organic waste, digest it and excrete simplified organic matter. They also assist the movement and involvement of poorly mobile microbial communities. Earthworms also fractionate the organic matter, thus allowing an increased surface area exposure to other decomposer microbes [137], and during the maturation stage, earthworms travel to freshly added undigested organic waste. This step is necessary to collect earthworm-free vermicompost while not disturbing the process. During the organic matter decomposition by earthworms, it has been demonstrated that pathogen removal occurs while transportation through the gastrointestinal track of the worm [138] and also during the process of composting. However, for complete pathogen removal and increased decomposition rate, combining of vermicomposting and composting is preferred. Similar to composting process, vermicomposting also produces greenhouse gases like CO₂ and N₂O depending on the raw material used [139]. The rate of decomposition, final product quality, worm mortality and greenhouse gas emission all depends on the vermicomposting parameters and therefore, the process parameters must be kept under control.

Zhou et al. [140] provided an overview for future waste management developments using various microbial cultures either individually or in the form of consortium, supplements, bulking agents, or their amalgamations for process improvement. Recent advancements in combining mathematical modelling, innovative supplements, and intelligence-assisted device designs with (vermi-) composting practices were highlighted. Maharjan et al. [141] assessed the suitability of vermicomposting for various types of OWs based on earthworm growth and reproduction as well as nutrient

content in final vermicompost from the papers published during the years between 2012 and 2021. *Eisenia fetida* was a commonly used species for vermicomposting and the majority of vermicomposting experiments were conducted on animal waste, followed by sewage and industrial sludge. Phuong [132] used three types of earthworms including *Eisenia fetida*, *Perionyx excavatus* and *Eudrilus eugeniae* (African nightcrawler) to prepare vermicompost from coffee waste. *E. fetida* worms were found to be the most suitable for processing coffee grounds. Mixed cultures of *E. fetida* and *Eudrilus eugeniae* resulted in improving treatment efficiency. It offered high efficiency, short processing times, labor cost savings and environmental benefits. In subsequent studies, Zhou et al. [142] reviewed the detailed mechanism of composting and vermicomposting using different methods, biochar, additives and bulking agents and highlighted the involvement of earthworms and microorganisms to the process of composting. The shortcomings and limitations of different technologies were compared with the traditional original compost technology.

4.3 Production of biochar from organic waste

Biochar is a compact solid attained after the thermo-chemical alteration of organic waste material in anaerobic environment [40, 143]. The Amazonian "Terra Preta" soils were first discovered in the nineteenth century by European explorers in the Amazonian region that contain large amounts of biochar derived carbon with greater microbial activity, better nutrient availability and crop production during exploration [82, 144, 145]. Recently, biochar has appeared as a potential supplement to soils as well as composts with many positive outcomes [42, 146, 147]. Generally, the carbon stored by plants via photosynthesis is returned to the atmosphere during natural decomposition of dead plants, thus completing the carbon cycle [25]. However, converting dead biomass into biochar disturbs the carbon cycle by storing carbon in stable form [40, 148].

Biochar is produced from various waste biomass sources, such as wood, manure, wheat or paddy straw, sewage sludge and municipal refuse [41, 149]. Biochar is produced by the method of pyrolysis under controlled conditions. The pyrolysis reactor's temperature typically varies from 300 to 1000 °C. Pyrolysis of plant biomass results in three streams of products: (i) noncondensable gases, (ii) a combustible bio-oil (liquid hydrocarbon) and water, (iii) biochar (solid byproduct) [150, 151]. The pyrolysis reaction can be either exothermic or endothermic based on the temperature of the reactants, but with decrease in reaction temperature, it becomes increasingly exothermic [152]. Also, the method should be energy-efficient and pyrolytic gases should be utilized within the procedure to reduce emissions [153]. Compared to pyrolysis at higher temperatures, instead of; pyrolysis at lower temperatures often results in a higher biochar yield and an improved volatile matter composition [154]. There are estimates of exothermicity or energy yield of 2.0 to 3.2 kJ g⁻¹ of biochar on slow pyrolysis [152]. Among different methods used for pyrolysis, microwave assisted pyrolysis is carried out at 300–500 °C ranging from minutes to hours, whereas slow pyrolysis is performed at 300–700 °C, and it takes hours [151]. At temperature range of 450–550 °C, pyrolysis hardly takes 1 min (fast pyrolysis) and flash pyrolysis is carried out 300–800 °C temperature within 1 s. Other pyrolysis methods include torrefaction (200–320 °C for hours), hydrothermal carbonisation (150–400 °C for minutes to hours) and gasification (> 800 °C for seconds to minutes). This allows regulation of physico-chemical properties of biochar.

Production of biochar also requires careful selection of feedstock and optimization of processing conditions to achieve maximum possible stability. Zhu et al. [155] stated that one of the most popular management techniques for achieving carbon neutrality in a circular economy is the pyrolysis of agro-residues into biochar. Agro-residues may be pyrolyzed to make biochar, which can then be used as an alternative energy source, a soil conditioner, and a replacement for activated carbon to create biochar that is more environmentally friendly and long-lasting. Krishnamoorthy et al. [156] reported that biochar properties were significantly affected by feedstock type and pyrolysis temperature. Sawdust biochar (lignocellulosic biomass) produced at 500°C had the highest struvite yield from urine (7.91 g L⁻¹), phosphate (97.9%) and ammonium recovery (87.1%), and relative crystal size (85.2%) compared to the non-seeded treatment due to its higher surface area, pore volume, and hydrophobicity of the biochar. Manure pellet biochar (non-lignocellulosic biomass) produced at 500°C also exhibited performance comparable to sawdust biochar produced at 500°C. Increasing pyrolysis temperature increased biochar's hydrophobicity, zeta potential, electrophoretic mobility and bulk density, irrespective of the feedstock type, thereby improving the seeding process. The ash content of biochar was negatively correlated with its surface area, pore volume, and particle size, but positively correlated with biochar's bulk density and suspension stability. In another study, Sun et al. [92] systematically investigated the effect of pyrolysis temperature on the structure of biochar prepared from crop straw of three crops (cotton, alfalfa, and wheat). A higher pyrolysis temperature led to a higher aromatization and graphitization degree of biochar and a numerical relationship between pyrolysis temperature and biochar surface properties (functional groups, carbonization degree) was established. Pyrolysis temperature controlled the content,

composition, and functional group evolution of biochar-derived dissolved organic matter. Although the amount of potentially toxic elements (PTEs) in biochar was concentrated after pyrolysis, the potentially risk of PTEs significantly decreased. The spin concentration of persistent free radicals in biochar prepared at 500 °C was the highest.

4.4 Organic waste utilization as phytostimulants

Organic phytostimulants or plant biostimulants are complex and heterogeneous organic substances that improve plant productivity and quality by activating different physiological and molecular processes in crop plants [157, 158]. Organic phytostimulants contain materials or microorganisms that may act as fertilizers and increase plant nutrition processes independently of the product's nutritional level and may also enhance one or more of the following traits of the plant and/or the rhizosphere of the plant, such as (i) nutrient usage efficiency, (ii) biotic stress tolerance/resistance, (iii) quality traits, or (iv) availability of restricted nutrients in the soil or rhizosphere [157, 159]. Phytostimulants can originate from diverse raw materials with different compositions that can significantly reduce the input of chemical fertilizers and agrochemicals in soils and hence, promote the environmental sustainability of agriculture. Their use as organic phytostimulants provides an interesting possibility for reducing the environmental stresses on agricultural systems [157]. These organic phytostimulants have been grouped into different families: seaweed extract, protein hydrolysates, humic and fulvic acid, chitosan, beneficial microorganisms and other complex phytostimulants [160–162]. However, the mechanisms of phytostimulants action are still being investigated. The main organic phytostimulants are summarized in this section.

4.4.1 Seaweeds extract

Seaweeds extract (SWE) or other plant extracts are either applied directly on soils or as foliar treatment in the form of purified products (such as the products of polysaccharides such laminarin, alginates, and carrageenans) [163, 164]. Micro- and macro-nutrients, N-containing chemicals (betaines), sterols, and phytohormones including auxins, cytokinins, and gibberellins are other components in seaweed extracts that support plant development [165]. The polysaccharides like laminarin, alginates, carrageenans, or their products may result in gel formation to enhance water retention. The polyanionic compounds in SWE helps in the fixation and exchange of certain cations and HMs for soil remediation. SWE may also involve some antioxidants or endogenous stress responsive genes regulators. Raja and Vidya [162] reported that seaweeds extracts contain bioactive compounds that activate several signaling pathways and defense-related genes/enzymes. Seaweed extracts and their components induce tolerance, mitigate stresses, can enhance nutrient efficiency and boost growth under stress conditions in agriculture.

4.4.2 Protein hydrolysates

Protein hydrolysates (PHs) consist of peptides or amino acids, which are derived from the enzymatic or chemical hydrolysis of proteins, and are derived either from plants or animals. PHs are recognized as biostimulants for sustainable crop production systems. PHs have been demonstrated to modulate N uptake and assimilation, stimulate shoot and root biomass resulting in enhanced plant growth and yield of various crops [166]. Protein hydrolysates obtained from animals acted as biostimulant for transplanted strawberry plants subjected to cold stress [167]. Some of the PHs may protect plants against HMs, improve soil microbial biomass and activities leading to increases in micronutrient mobility and soil fertility, which might be indirectly enhancing nutrient uptake and plant biomass [168]. Compounds such as glycine, betaine and proline derived from PHs confer antioxidant activity and contribute to the mitigation of various abiotic stresses [169]. However, the effectiveness of different PHs is dependent on various factors, including source, production process and plant species.

4.4.3 Chitosan

Chitosan is a biodegradable and biocompatible polysaccharide obtained from partial chemical or enzymatic deacetylation of chitin. It is the second most common natural polysaccharide after cellulose. Due to its natural origin and multidirectional bioactivity of this polycationic polymer, chitosan is being used as a biostimulant, a plant protection product, an elicitor, or an agent to increase the storage stability of plant raw materials [170]. Chitosan has antifungal, antibacterial, antiviral, and antioxidant activity, and is involved in induction of plant defense mechanisms and regulation of metabolic processes. The purified preparations obtained from brown algae such as laminarin and chitosan-based nanomaterials

are extensively used in agricultural and biomedical applications [171]. The binding of chitosan to cell receptors with lower specificity causes accumulation of H_2O_2 and leakage of Ca^{2+} into the cell, which results into stress signalling and expression of growth regulatory mechanisms [172]. Chitosan or chitosan-based nanomaterials suppresses plant diseases caused by fungal pathogens and play important role in mitigation of abiotic stresses such as drought, cold and salinity; thereby leading to improvement of plant growth and increases in agricultural productivity [171, 173]. The effectiveness of chitosan interactions is determined by its origin, deacetylation degree and acetylation pattern, molecular weight, type of chemical modifications, pH, concentration, and solubility. Therefore, extensive studies are needed to evaluate the bioactivity and toxicity of chitosan nanoparticles and chitosan conjugates for their large-scale use in different crops under field conditions.

4.4.4 Complex biostimulants

Complex biostimulants include non-pathogenic bacteria, mycorrhizal fungi and other beneficial microorganisms, which inhabit the soil, rhizosphere, as endophytes or in phyllosphere [174–176]. These microbes improve soil fertility, and help the plants by improving their nutrient uptake, by mitigating abiotic stresses and may prevent invasion of pathogens and pests [160, 177–179]. Many microbial biostimulants increase crop yield and quality under pot house and field conditions [180–182]. Some antagonistic rhizospheric bacteria such as *Pseudomonas* and *Bacillus*, and fungal endophytes like *Trichoderma* spp., *Sebacinales*, *Phanerocheate* etc. are being used as biocontrol agents and as biopesticides [147, 183, 184]. The symbiotic association of arbuscular mycorrhizal fungi (AMF) with crop plants is widely recommended to promote sustainable agriculture [185, 186]. AMF treatment helps in absorption of macro- and micro-nutrients by plants, in maintaining water balance and mitigating biotic and abiotic stresses [187, 188]. The tripartite associations between mycorrhizal fungi, plant and rhizobacteria along with application of compost have been reported to improve soil quality and increase plant health [189, 190].

Inoculation with mutualistic endosymbionts like *Rhizobium* and plant growth promoting rhizobacteria (PGPR) is mostly favoured as biostimulants [191, 192]. Single or dual microbial inoculation was found as an efficient biostimulant in legumes with multifunctional growth promoting traits [193]. For instance, coinoculation of chickpea with *Mesorhizobium* sp. Cicer strain Ca181 and *Pseudomonas* strains MRS23 and CRP55b, increased the nodule formation by 68.2–115.4% on chickpea at 80 and 100 days after planting under sterile conditions in comparison to single inoculation of *Mesorhizobium* strain [194]. Shoot dry weight ratios of coinoculated treatments at different stages of plant growth varied from 1.18 to 1.35 times that of *Mesorhizobium*-inoculated and 3.25 to 4.06 times those of uninoculated plants. In similar studies, Sindhu et al. [195] reported that coinoculation of *Bradyrhizobium* strain S24 with *Bacillus* strains in green gram (*Vigna radiata*) caused 1.28 to 3.55-fold enhancement in shoot dry mass at 40 days of plant growth. Coinoculation of *Bradyrhizobium* strain with majority of *Bacillus* strains promoted the nodule number on green gram roots and also increased expression of nitrogenase activity under pot house conditions. Under salt stress conditions, separate inoculations of *Rhizobium* sp. LSMR-32 and *Enterococcus mundtii* LSMRS-3 improved sprouting of seeds, seed harvest, crop's physical characters like height and biomass, chlorophyll concentration, and nutrient assimilation as compared to uninoculated crops [196]. Inoculation with both these strains also enhanced the quantity of leghemoglobin, nodule number, and nodule biomass along with phosphatase and dehydrogenase enzyme levels of soil. The link between the microbiome and plant health has recently been reviewed [197] and microbiome applications showed significant impacts on plant development and sustainable crop production.

4.5 Production of biofuels and chemicals from lignocellulosic biomass

Rapid population expansion and advancement in industrialization and modernization has led to a hike in the demand of energy in recent years [198]. This has resulted in the depletion of conventional energy resources such as oil, natural gas and coal. On the other hand, the use of fossil fuels has caused drastic pollution of air and water resources, along with increase in global temperature and ozone layer depletion [2, 199, 200]. Recently, utilization of agricultural waste for the production of energy has emerged as innovative technology for replacement of fossil fuels in the production of sustainable energy [198, 201, 202].

Crop residues primarily consist of lignocellulosic biomass, which is widely available as a low-cost renewable feed-stock. Nearly half the quantity of rice-husk produced in India is used as animal-fodder and the other half portion is discarded in the environment or burnt by the poor farmers. Therefore, appropriate technology may be developed for conversion of rice-husk into useful products after value addition including electricity and biofuel. Recently, many studies

have demonstrated that lignocellulosic biomass including seaweeds may also be used for the production of biofuel (biogas, bioethanol, biodiesel, biohydrogen) and bioenergy to resolve the issues of fossil fuel scarcity and climate change [39, 94, 202, 203]. However, to utilize lignocellulosic biomass for the purpose of producing sustainable energy, it must be subjected to pretreatment to speed up its fragmentation and reduce lignin concentration.

Deivayanai et al. [88] discussed various pretreatment techniques to expand the dissolution capacity and surface area of lignocellulosic waste material. Operational parameters, yield percentages, and a techno-economic analysis were studied for the biological conversion of lignocellulosic waste to hydrogen. In another study, enhanced hydrogen production was achieved via integrated lignocellulosic biomass bioprocessing employing a nanobiotechnology method [204]. Utilizing raw corn cob as a source of lignin, peroxidase like CeFe_3O_4 nanoparticles (NPs, 4.0 g L^{-1}) were utilized to break down the lignin in order to produce cellulose-hemicellulose portions for fermentation with *Clostridium cellulovorans*. Using nanoparticles-treated corn cob biomass, fermentation was carried out with *C. cellulovorans* strain, and it generated a maximum 78.45 mL of collective hydrogen.

In addition to energy, various enzymes such as xylanase, pectinase, ligninases, amylases and other enzymes can also be produced from rice-husk. Li et al. [205] reported that sugarcane bagasse (a byproduct of sugarcane industry) may be used in the production of hydrophobic or water resistance gel, carbon fibre, bioactive natural products, bioplastics, thermoplastic, foams, coatings and thermosetting materials with broad industrial uses. In addition, agricultural waste is also used in the paper-pulp and cardboard industries. The pulp of wood is primarily used for production of cardboards and very low amount (only 4%) of the total crop waste is utilized. Thus, wide range of value-added products such as bio-coal, biofertilizers, bio-bricks, bio-plastics, paper, biofuels, bioelectricity, chemicals, organic acids and industrial enzymes are currently being produced from lignocellulosic waste, which has large impact on the economy [8, 202, 204].

5 Influence of organic amendments on different biological processes in soil

For providing sufficient food to exponentially increasing human population, intensive agriculture practices are followed by farmers worldwide, which generates significant amounts of organic agricultural waste [8]. For processing of organic waste, various biological technologies such as composting, anaerobic digestion, vermicomposting and biochar production are preferentially employed [89, 110, 113]. These sustainable and eco-friendly technologies ensure food security along with mitigation of pollution problems. For instance, the amendment of compost, vermicompost and biochar to the soil has been found to increase soil fertility, stimulation of beneficial microbial community, mitigation of HM stress, suppression of soilborne pathogens and detoxification of pesticides leading to increased plant growth and crop harvest along with less dependence on chemical fertilizers [28, 32, 147, 206] (Fig. 3). Massaccesi et al. [62] reported that the biomass of lettuce (*Lactuca sativa* L.) plants was increased from 2- to 2.5-fold in all compost and biochar amended soils in comparison to the control after 45 days of cultivation.

5.1 Increase in nutrient availability in soil

Optimal environment, sufficient moisture, and the right quantity of nutrients are essential for the development of crop plants. But, the soil's potential to deliver nutrients is diminishing with time due to soil erosion, land degradation and intensive crop production systems [57, 207]. For enhancing crop productivity, farmers use excessive amount of inorganic fertilizers, which have deleterious effects on soil health leading to unsustainable yields [208], and also cause environmental pollution leading to public health hazards [6]. Continuous use of chemical fertilizers in the field also affects the proliferation of beneficial microbial community in the soil [209]. In order to increase crop yield, it is necessary to upgrade the nutrient distribution system in a sustainable way to provide adequate food to exponentially rising human population [33, 210, 211]. The effect of different types of organic amendments (OAs) (i.e., unamended control, conventional farmyard manure (FYM), kraal manure, improved FYM, compost, vermicompost, urban waste, taroka, poultry droppings, pig manure) and earthworm *Pontoscolex corethrurus* (Rhino drilidae) inoculation was assessed on the growth and nutrition of rice [135]. The poultry droppings had the best fertilizer performance, followed by improved farmyard manure and pig manure in treatments without earthworms after eight weeks of growth. Recently, application of processed products obtained from organic waste materials during composting, vermicomposting, biochar and phytostimulants has emerged as novel agri-technologies for preparing nutrient rich fertilizers [32, 35, 106, 212].

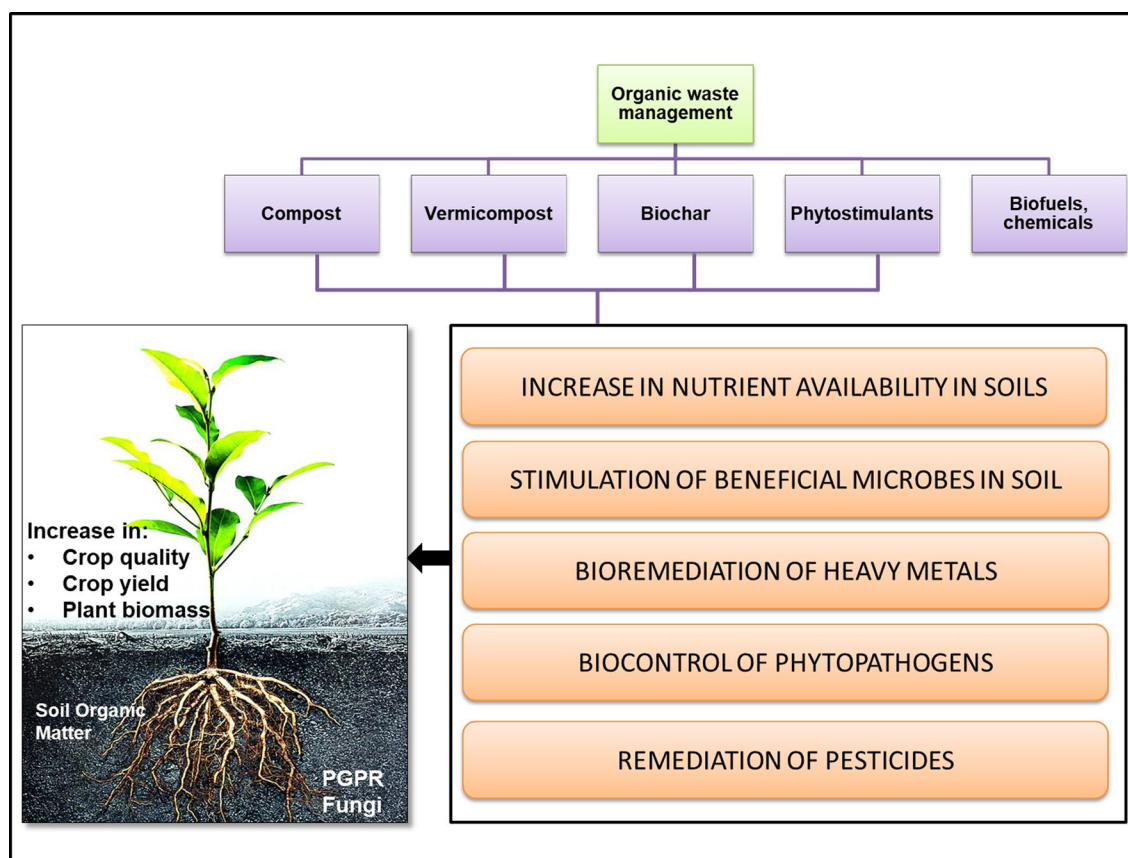


Fig. 3 Various agri-technologies used to manage organic waste and their role in functioning of various biological processes and plant growth promotion

5.1.1 Enhancement of nutrient availability by composting

Addition of compost in the farm soil increases nutrient levels and improves soil structure [9]. It also improves soil fertility, porosity and soil microbial biomass [213]. A good quality compost is an excellent source of three macronutrients for the plant i.e., nitrogen, phosphorous and potassium (NPK). Although, as compared to inorganic fertilizers, their concentrations in compost is low, but their regular application may fulfill the plants nutrients need, and hence, reduce the application of chemical fertilizers. Also, addition of compost in soils can benefit by altering the soil's heavy metal bioavailability and can mitigate their toxic effects on animals and plants due to different processes such as adsorption, precipitation and redox reactions.

The leachates obtained from the composting process (used as raw or diluted form) were found to serve as substrate for utilization by microbes involved in composting process in shake-flask fermentation assays. For example, in comparison to raw leachate, *Bacillus* sp. thrived better in diluted leachate [214]. Amendments of leachates with yeast and whey permeate improved the growth of *Bacillus* sp. and also the sporulation rate. Co-substrates produced the greatest sporulation rate, causing over 89% of vegetative cells to produce spores. Results indicated that leachate valorization approach will reduce the volume of leachate waste for treatment and will also help the circular economy produce biological amendments at a reasonable price. In another study, improvement in biological phosphorus transformation and nutrient content in composting products was observed by inoculation of kitchen waste with phosphorus-solubilizing bacteria (PSB) [215]. Treatment with PSB considerably improved enzyme activity and enhanced the levels of bacterial diversity, total P, Olsen P, citric acid P, and organic matter (OM) decomposition. In a recent study, Priya et al. [216] explored the application of soil amendments compost (CMP) and gypsum, and signaling compounds such as strigolactones (SL), salicylic acid (SA), SL and SA or SL-SA, and coumarins to modulate symbiotic root-microbe interactions for improving legume productivity in saline soils. The SL-SA treatment outperformed CMP in enhancing nodulation and AMF colonization. Only CMP induced a significant shift in the rhizosphere and root microbiome structure. Increased abundance of AMF *Rhizophagus* in the rhizosphere and AMF *Diversispora* in the

root endosphere of SLSA treatment, and *Actinobacteria* as well as *Streptomyces* in the root endosphere of the SLSA treatment was observed.

5.1.2 Contribution of vermicomposting towards enhancement of nutrient availability

Soil amendment with vermicompost increased the amount of stored carbon under rice-rice cropping pattern [217] and promoted root growth as well as ginsenoside content of *Panax ginseng* [218]. Supplementation of vermicompost with recommended dose of inorganic fertilizers was found to enhance N, P and K content in soil in comparison to using only inorganic fertilizers. Sahariah et al. [217] studied soil physical parameters during vermicomposting process and it was observed that addition of vermicompost in the soil improved its water holding capability and decreased bulk density, in comparison to the addition of NPK alone or NPK with farmyard manure. Addition of vermicompost also enhanced microbial biomass as well as their enzyme activities (dehydrogenase, urease, phosphatase and aryl sulfatase) in soil [219]. Increased microbial biomass and dehydrogenase activity was positively correlated with increased ammonical N, nitrate N and orthophosphates. These factors increased nutrient cycling rates in soil and promoted synthesis of plant growth regulating substances.

5.1.3 Enhancement of nutrient availability by biochar application

Biochar amendment in soil results in increased nutrient cycling and improvement in soil fertility, water retention capacity and enhanced microbial activities and also influenced soil characteristics [42, 146, 220, 221]. Different biotic and environmental stresses in soil were found to affect the mineralization rate of biochar [222]. Biochar addition to soil has also been reported to reduce greenhouse gas emissions from soils [148, 223]. The biological, chemical, and physical characteristics of soil can all be considerably enhanced by biochar [221], including soil pH, nutrient availability, cation-exchange capacity (chemical properties); bulk density, porosity and water retention capacity (physical properties); and improvement in microbial population, variety and activity of soil microbes (biological properties). The results of biochar application on soil depends on different variables such as type of feedstock used, conditions at the time of biochar production, the type of soil to which biochar is applied, together with the quantity of biochar used [42, 222].

Amendment of biochar resulted in improved soil fertility and sequestration of carbon and also increased crop yield, alleviated plant stresses, immobilized HMs and reduced their uptake by plants in soil contaminated with organic pollutants [40, 224]. Application of both compost and biochar to soil further enhances its agronomic values as well as results in minimum nutrient loss, which can potentially replace nitrogen fertilizers [225]. Addition of biochar to composts improved its properties and biochar amendment improved aeration in compost because of low density and high porosity of biochar, thereby increasing the humification process and overall microbial activities [226]. Also, because of high sorption ability of biochar, it may reduce excess N loss from compost. Biochar can also immobilize organic pollutants and HMs during composting and can lower emissions of greenhouse gases [224, 227]. Bouhia et al. [228] reported that addition of biochar singly or in biochar-compost combination (BCC) in *Zea mays* grown in pots filled with alkaline soil increased soil TOC, EC, and pH under greenhouse conditions. Furthermore, adding biochar and compost increased the levels of macro- and micronutrients compared to those under single application of biochar. The soil fertility improved significantly with regard to available P and K, N, and micronutrients. Although, single application of biochar had a negative impact on mycorrhizal symbiosis but single application of compost gave the best results in terms of plant growth and soil fertility improvement.

Xie et al. [189] carried out four treatments in ginger field soil i.e., control (no compost or PGPR; CK), compost (TC), compost + *Bacillus subtilis* (TC-BS), and compost + *Bacillus amyloliquefaciens* SQR9 (TC-BA). According to the findings, TC considerably enhanced soil organic matter content by 17.34%, alkali hydrolysable nitrogen by 21.66%, available P by 19.56%, and accessible K by 37.20% when compared to the control (CK). Neutral phosphatase, sucrase, and urease activity in the soil all rose by 55.89, 35.59, and 57.21%, respectively. According to research by Ali et al. [229], initial reductive soil disinfestation (RSD) treatment, biochar amendment (RSD-BC), and collective applications of microbes (such as *Trichoderma* and *Bacillus* spp.) and biochar (RSD-SQR-T37-BC) increased soil pH, soil organic carbon, and ammonium in the treated soils while lowering nitrate concentration. Li et al. [109] provided an overview of biochar application for organic waste co-composting and its biochemical transformation mechanism. As an amendment in composting, biochar worked in the adsorption of nutrients, the retention of oxygen and water, and the promotion of electron transfer. Biochar addition caused changes in community structure, and mediated metabolic activities involved organic matter degradation. The participation of biochar enriched the α -diversity of microbial communities at all stages of composting,

and ultimately reflects the high γ -diversity. However, the mechanism of biochar action on composting microbes at the microscopic level needs further in depth studies.

In another experiment, Hu et al. [43] reported that addition of biochar increased the soil's ammonium N level by 8.3–101.5%, and the total N content by 4.7–32.3%, respectively. The use of biochar also drastically altered the overall composition of the soil's bacterial community, and the N content of the microbial biomass rose with the addition of various doses of biochar. With amendment of 2% biochar, the expression of the genes encoding nitric oxide reductase and nitrous oxide reductase rose by 16.4% and 16.0%, respectively compared to the control treatment. Recent approaches to utilize solid/liquid digestate, nutrient retrieval, microalgae culture, and mixing with biological and thermochemical processes were described by Wang et al. [230]. The unique methods for evaluating anaerobic digestate were put forth, together with their difficulties and potential. Because of the possible commercial value of goods obtained from hydrochar/biochar, integrating anaerobic digestion with thermochemical processes like hydrothermal carbonization may be an attractive strategy.

In order to realize the idea of circular economy and maximize the advantages of digestate valorization, the use of digestate has steadily changed from being seen as waste to products. For instance, Xie et al. [105] showed that treatments of lemon plants with chemical fertilizer (CF) and chemical fertilizer combined with biochar (CFBC) significantly increased potential nitrification rates (PNR) by reducing denitrification potential in a pot experiment. On the other hand, treatments with manure (M) and MBC (a combination of manure and biochar) increased the denitrification potential by 2192.7 and 1989.9%, respectively in comparison with CK (no fertilization) treatment. Treatments of lemon plants with CF and CFBC increased ammonia-oxidizing archaea (AOA) and *amoA* gene (encoding ammonia monooxygenase) copies from 40.2 to 101.7%, whereas treatments with M, MBC and FMBC (combination of fertilizer, manure and biochar) reduced them from 66.5 to 81.9%. Ammonia-oxidizing bacteria (AOB) and *amoA* gene copies decreased by 33.1–81.9% after five different fertilization treatments. Sifton et al. [206] examined the effects of two wood biochar types with inoculation of *Bacillus velezensis* and an inactivated yeast (IY) biofertilizer on *Acer saccharinum* (silver maple) saplings in greenhouse experiment. All treatments combining biochars with biofertilizers significantly increased sapling growth and biomass (91%) compared to control treatments. Nutrient uptake patterns and nutrient vector analyses indicated that combined biochar and biofertilizer treatments effectively addressed nutrient limitations of both macronutrients (N, P, K, Mg, Ca), and micronutrients (B, Fe, Mn, Mo, Na, S, and Zn). Biochar-biofertilizer treatments also reduced foliar concentrations of Cu, suggesting potential to mitigate toxic metal impacts common in urban forestry.

In a recent study, Zhao et al. [231] reviewed the effects and potential mechanisms of biochar on microorganisms in anaerobic digestion (AD) systems and found that biochar addition can provide habitats for microbial colonization, alleviate toxins stress, supply essential nutrients, and accelerate interspecies electron transferring. Moreover, it also improves microbial community diversity, facilitates exopolysaccharides secretion, enhances functional enzyme activity, promotes functional genes expression, and inhibits the antibiotic resistance genes transformation. Biochar targeted design, in-depth microbial mechanisms revelation, and modified model development were suggested to promote the application of biochar-amended AD technology. In another experiment, the long-term effects of biochar application were studied on soil physicochemical properties, salt concentration, nutrient availability, soil enzyme activities, and rice yield under highly saline-alkali paddy soils [212]. The results showed that one-off application of biochar continuously reduced the soil Na^+ concentration, Na^+/K^+ ratio, $\text{Na}^+/\text{Ca}^{2+}$ ratio, exchangeable sodium percentage (ESP), and sodium adsorption ratio. It enhanced the concentration of K^+ , Ca^{2+} , Mg^{2+} , and cation exchange capacity (CEC) over the 6-year study, indicating its longer-term positive impact. Furthermore, the one-off biochar application, especially under high application rate treatments (i.e., 3.0% biochar application (T3.0), and 4.5% (T4.5), significantly and continuously improved nutrient availability and soil enzyme activities. The grain yield of 1.5% biochar application (T1.5, T3.0, and T4.5) surpassed that of control untreated (T0) by 116.38%, 141.24%, and 145.20%, respectively.

5.1.4 Enhancement of nutrient availability and soil fertility by biogas slurry

The biogas slurry discharged from the common anaerobic digesters (biogas plants) is used as organic manure to improve nutrient availability and soil fertility. The cattle dung after passing through the gas plant becomes rich manure and it contains 2% nitrogen as against 0.75% in ordinary compost manure. The domestic biogas digester slurry could also be used in the production of vermicompost [232]. The residual manure of biogas plant helps to meet the fertilizer requirements of farmers in a more economical and efficient manner. Thus, application of biogas slurry offers an alternative to chemical fertilizers and it offers a promising approach to a biogas-linked agroecosystem that restores sustainable coordination between cropping and animal husbandry under an intensive agriculture production system. It helps in

checking of fertilizer shortage, reduces the pollution and helps in improvement of hygiene, sanitation and better health for people in rural areas. However, the responses of soil fertility and microbial communities to long term use of biogas slurry need to be explored in different soil types and regions in realizing ecologically recycling agriculture.

Tang et al. [114] applied biogas slurry (BS), chemical fertilizer (CF) or substitution of half the chemical fertilizer with biogas slurry (BSCF), and a control treatment (without fertilizer addition) to a fluvo-aquic soil. Organic carbon and available nutrient concentrations (N, P and K) were enhanced in all fertilization treatments relative to the control. Among different fertilization treatments, BSCF resulted in the greatest diversity and most evenly balanced assemblages of both bacteria and fungi at the phylum level. Half substitution of the chemical fertilizer by biogas slurry gave the greatest improvement in soil structure and nutrient availability, and this was associated with greater microbial diversity and better balanced microbial communities. In another study, the effects of limed manure-anaerobic digestate as a nitrogen source for plant growth and on N_2O emission was examined and compared with compost [233]. The combination of compost application and liquid-N fertilizer resulted in surface N_2O fluxes over $0.7 \text{ mg m}^{-2} \text{ d}^{-1}$, which was correlated with ammonium concentration in the soil. The presence of N_2O in the rhizosphere was only detected in compost-amended soil 2–10 days after plantation. A significantly-lower surface N_2O flux of $0.4 \text{ mg m}^{-2} \text{ d}^{-1}$ was recorded with application of limed-digestate, probably due to its effects on nitrogen-transforming microorganisms. Both compost and limed-digestate enhanced plant growth. The results demonstrated that limed-digestate can be applied to agricultural soil as an efficient substitute for compost as it effectively supports plant growth with substantially-lower N_2O emissions.

5.2 Stimulation of beneficial microbes in soil

Specific microorganisms including bacteria, fungi, algae, actinobacteria, archaea, protozoa, and nematodes inhabit the soil and rhizosphere, and influence the development and growth of crop plants [33, 182, 234, 235]. The interactions of surrounding microbes with plants are controlled by exchange of signalling molecules in their environment [236, 237]. The narrow area in close contact with the surface of root, termed rhizosphere, is most affected area by microbial activities [238]. In the rhizosphere, plant roots secrete a wide spectrum of metabolites in root exudates ranging from organic acids, sugars, alcohols, amino acids, nucleotides, and secondary metabolites [239, 240]. These exuded plant metabolites play important role in fostering beneficial interactions with the surrounding microbial community in the plants' rhizosphere and modulate the dynamics of plant–microbe interactions in fulfilling plant niche-based requirements [241, 242].

It has been estimated that in general bulk soil contains 10^7 to 10^9 bacterial cells gram^{-1} and there is almost 10 to 100-fold more bacteria in the rhizosphere [243, 244]. Approximately, 30,000 prokaryotic species inhabiting the soil have been found to impact the development and productivity of plants [240, 245]. The configuration of rhizosphere microbiota prominently affects the communications between plants and soil, thus helping plants in adapting towards adverse environmental conditions including biotic and abiotic stresses [178, 246, 247]. The soil amended with low concentrations (at 0.01% and 0.1%) of pyroligneous acid (PA; obtained from pyrolysis of lignocellulosic raw material) increased microbial diversity and dehydrogenase activity in comparison to the control treatment and high concentrations (1% and 5%) of PA amendment to soil [248]. Interestingly, population of the plant growth promoting bacteria (PGPB) i.e., *Azospirillum*, *Acetobacter*, *Beijerinckia*, *Bradyrhizobium*, *Herbaspirillum*, *Mesorhizobium*, *Nitrosomonas*, *Pseudomonas* and, *Rhizobium* was stimulated at lower concentrations of PA.

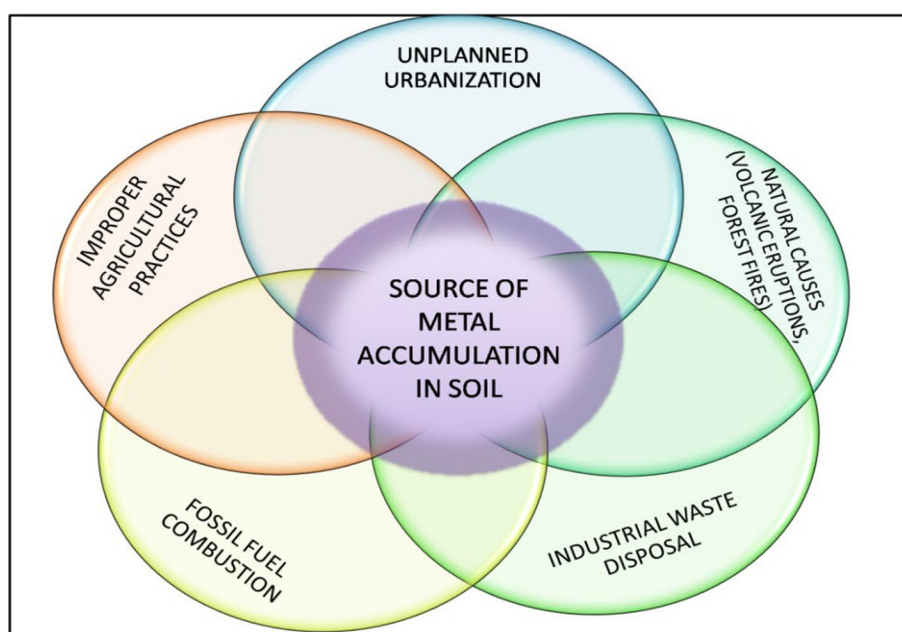
The application of phosphorus fertilizer enriched with microorganisms including *Paenibacillus polymyxa* strain CHT114AB, *B. amyloliquefaciens* strain AF75BB and *Bacillus* sp. strain CZP4/4 was found to improve enzymatic activity and the genetic and functional diversity of the soil microbial communities [249]. The addition of compost and inoculation of beneficial microbes preferentially stimulated the specific microflora in the soil. For instance, ginger field soil was treated with “compost (TC), compost + *Bacillus subtilis* (TC-BS), and compost + *Bacillus amyloliquefaciens* SQR9 (TC-BA)” [189]. TC-BS treatment significantly upgraded soil bacterial variety than control treatment i.e., no compost or PGPR “(CK) and TC, and relative abundance of beneficial Proteobacteria, Acidobacteria, Chloroflexi, and *Bacillus*” microorganisms dominated. This study demonstrated that *B. subtilis* inoculation increased the variety of bacteria, changed the bacterial population framework, boosted advantageous microorganisms, and encouraged a flourishing rhizosphere in soil from a ginger field treated with tomato compost. In subsequent studies, five fertilization treatments i.e., “chemical fertilizer (CF) and chemical fertilizer combined with biochar (CFBC), manure (M), MBC (manure combined with biochar and FMBC)” decreased the ammonia-oxidizing bacteria (AOB) by 33.1–81.9% [105]. The redundancy analysis demonstrated that “pH, available P, carbon to nitrogen (C/N) ratio, soil moisture content, nitrate nitrogen, and ammonium nitrogen” present in the rhizosphere influence the population structure of ammonia-oxidizing archaea and ammonia-oxidizing bacteria.

Castellano-Hinojosa et al. [250] examined the effect of four citrus rootstocks and compost on the diversity, composition, and predicted functions of active rhizosphere bacterial communities. Variations in the active bacterial rhizobiome were strongly linked to root nutrient cycling (C, N, and tryptophan metabolisms) and specific rootstocks, particularly in soils treated with compost. Direct positive relationships between enriched taxa in treated soils and specific root nutrients were detected. Liu et al. [251] investigated the impacts of adding FeSO_4 and biochar to cattle manure and rice straw composts on functional genes controlling N loss, bacterial community, nitrification, and denitrification. Four treatments were made including a control group (CP), and CP + 4% biochar (TG1), 4% FeSO_4 (TG2), or 2% FeSO_4 and 2% biochar (TG3). In comparison to control, treatments TG1–3 showed lower total nitrogen loss rate, and TG3 resulted in reduced NH_3 (52.4%) and N_2O (35.6%) emissions to mitigate nitrogen loss. The abundance of *amoA* (encoding ammonia monooxygenase) and *narG* gene (encoding membrane-bound nitrate reductase) in TG3 was higher than in the other groups, and TG3 was beneficial to the growth of Proteobacteria and Actinobacteria. According to redundancy and Pearson analysis, TG3 had a positive effect on the nitrification process by increasing the abundance of *amoA* and *narG*. Li et al. [252] performed high-throughput sequencing of soil microbial community structure after treatment of cucumber with 5% (w/w) sweet pepper straw biochar (SPSB) concentration and observed that this biochar concentration significantly altered soil microbial community structure and enhanced bacterial and fungal diversity. It increased beneficial bacterial groups (*Firmicutes*, *Actinobacteria* and *Bacillus*) and modified fungal communities, with a decrease in *Ascomycota* and *Aspergillus* and shifts in *Penicillium* abundance. Functional genomic analysis indicated enrichment in bacterial metabolic pathways and fungal replication and expression genes.

5.3 Bioremediation of heavy metals

Heavy metals such as lead, arsenic, mercury, copper, nickel, cadmium and zinc are being accumulated in soils due to various natural (volcanic eruptions and forest fires) as well as anthropogenic activities like application of chemical fertilizers, sewage sludge disposal, release of industrial waste, power stations emissions, fuel combustion by automobiles etc. (Fig. 4). Although, some of these metals are essential for plant growth in trace amounts, but their excess concentrations can poison the soil and negatively affect plant growth, physiology, metabolism and crop production. In soil, these HMs are non-biodegradable [253]. The physical and chemical treatment of HM remediation in soils is a laborious, expensive and non-sustainable process. Therefore, scientists are engaged in developing technology to sustainably remediate HMs from agronomic soils by using different metal tolerant microorganisms as bioremediation agents [18, 21]. Furthermore, mitigation of heavy metal pollution in soils through addition of compost, vermicompost and biochar has gained considerable attention because of its sustainability, low cost and health benefits [134, 254].

Fig. 4 Graphical representation of different sources for adding heavy metals in soil



Under the abiotic stress caused due to accumulation of HMs, growth of soil microbiota and their functioning in different ecological processes is negatively affected. Bioremediation involves the use of microorganisms to clean the polluted environment by detoxifying different types of pollutants [21, 255, 256]. These soil microbes employ a variety of mechanisms to survive under conditions of metal toxicity. They can store, alter, modify or detoxify HMs by binding, immobilizing, oxidizing and volatilizing them [257, 258]. The basic methods of bioremediation of soil pollutants by microorganisms include biosorption, immobilization, bioaugmentation and biotransformation (Table 1). Generally, two types of bioremediation i.e., in situ and ex situ methods have been adopted by metal tolerant microbes for successful remediation of HMs from soil. During in situ method, genetic engineering techniques are used, while in the ex situ method, the contaminated soil is removed from its original site, moisturized and amended with nutrients [259] Some common examples of metal-tolerant microorganisms include *Bacillus*, *Pseudomonas aeruginosa* (Zn and Cu), *Citrobacter* spp. and *Chlorella vulgaris* (Cd, Pb, U, Cu, Au, Ni, Hg), *Aspergillus niger* (Cd, U, Zn), and *Rhizopus arrhizus* (Cd, Ca, Hg, Pb, Ag) [260].

Application of compost and biochar in soil also contributed in mitigation of HMs stress using physico-chemical approach in the HM polluted soil [104]. The stoichiometric interactions between different microbial enzymes such as N-acetylglucosaminidase, β -glucosidase, leucine aminopeptidase and acid phosphatase were examined after application of heavy metals. Application of biochar to HM-polluted soils enhanced the activity of acid phosphatase and inhibited activities of other enzymes. In addition, application of biochar-compost strengthened all these enzymes activities, which were inhibited due to HMs. However, the stoichiometric ratio of C/P, C/N and N/P were changed due to application of the biochar and compost. Along with alleviating the phosphorous-limitation in soils under copper stress, biochar also alleviated the carbon-limitation in soils under Cd, As and Zn stress. Gao et al. [261] reviewed the recent advances in bioaugmentation technology for efficiently removing poisonous contaminants from organic waste contaminated soils by using microbes or their enzymes and bioremediation additives. A thorough overview of several bioaugmentation techniques, including bacteria, fungi, their enzymes, and some other bioremediation additives, was provided. The long-term stability of wide-range bioremediation carriers, microbial/enzyme agents targeting coupled pollutants, desorption of ecologically beneficial additives, and minute molecular biological stimulants were considered, while evaluating the future of bioaugmentation.

Li et al. [262] used vinegar-residue biochar for remediation of heavy metal cadmium from contaminated water and soil. Rwiza et al. [263] pyrolyzed lignocellulosic materials i.e., rice husks and maize hulls, which were burned at 500 °C to create biochar, and were subsequently activated with KOH and ZnCl_2 as activating agents. Compared to the raw hulls and husk, the modified biosorbents absorbed Pb^{2+} more quickly and effectively. In comparison to the sorption capacity of 15 mg g⁻¹ achieved by raw maize husk (CH), the KOH-treated maize husk biochar (KOH-CHB) acquired a removal capacity of > 97 mg g⁻¹ in the first 15 min. By pyrolyzing cannabis and cinnamon at various temperatures, Omidid et al. [264] produced biochars that functioned as potential adsorbents. The Cd (II) adsorption onto biochars was successful at pH 7.0 and at a temperature of 65 °C. A substantially shorter contact time and high Pb^{2+} removal by activated biochars were likely caused by improved structures. The low-cost materials used in the current study showed the potential to remove harmful Pb^{2+} from water at a reasonable price.

Foong et al. [265] presented a thorough examination of the use of biochar made from rice straw (RSB) to treat wastewater and remove various sorts of impurities. While magnetic biochar and electrochemical deposition have become promising methods for biochar modification, acid-modified RSB has the ability to eliminate metal ions and organic molecules. In another experiment, the highest relative reductions of HMs i.e., Pb, Cu, Cd, and Zn concentrations were achieved in shoot of Chinese mustard (*Brassica juncea* Czern) at the level of 74.72, 69.19, 54.16, and 53.86, respectively,

Table 1 Different bioremediation methods and their salient features

Bioremediation technique	Features
Biosorption	• Carried out by extracellular exopolysaccharides (EPSs) and polymeric substances linked with bacterial cell walls • They function by micro-precipitation of metals and proton exchange • Biosorbants include (i) EPSs, (ii) living microbial cultures and (iii) dead biomass
Immobilization	• Reducing heavy metal mobilization from polluted sites by manipulating their physical state or chemical characters
Bioaugmentation	• Introducing specific microorganisms at a polluted site to consume particular pollutant • Affordable, more effective and fast process
Biotransformation	• Heavy metal–organic complexes are changed into a precipitated, water-soluble state that is less hazardous

by combined application of biochar and salicylic acid, as compared with control [45]. The combined treatments of rice husk biochar (RH; 0, 2.5, and 5%) and foliar application of salicylic acid (SA; 0, 0.5, and 1 mM conc.) at a high rate, also caused highest increase in fresh (19-fold) and dry (three-fold) shoot weight of Chinese mustard. Interestingly, combined application also significantly improved antioxidant enzymes and increased chlorophyll a and b and carotenoid contents by approximately 300, 400 and 250%, respectively in comparison with the untreated soil. Contents of H_2O_2 , malondialdehyde, and proline were significantly stimulated in the HM-contaminated soil, whereas amendment with RH alone or with sprayed SA significantly reduced their content. In similar studies, Zhang et al. [266] isolated *Klebsiella variicola* with exceptional P solubilization capability (155.30 mg L^{-1} at 48 h) and Cd adsorption rate (90.84% with 10 mg L^{-1} Cd primary conc.). The P content and biomass of pakchoi (*Brassica chinensis* L.) inoculated with *K. variicola* and treated with phosphoric acid and ball milling co-modified biochar (PBC), were considerably enhanced, though the Cd content in the leaves and roots of pakchoi was lowered by 25.90–43.46%. In addition, the collective application promoted the conversion of the resistant P fractions to bioavailable P and aided in the immobilization of the 20.12% exchangeable Cd into reducible, oxidizable, and residual Cd in the soil. Rajput et al. [254] demonstrated that application of biochar at 2.5% level in soils along with metal-tolerant bacteria reduced the Zn uptake of roots by 44 and 20%, in comparison to polluted soil having zinc oxide nanoparticles.

5.4 Biocontrol of phytopathogens

Plant pathogens coupled with drastic climate changes are the major constraints for improving crop productivity and providing healthy food for ever-increasing human population. Various pathogenic microorganisms cause different diseases on major crops affecting up to 20–40% annual losses in crop yields globally, which seriously jeopardize food security and the quality of agricultural production [267, 268]. Farmers use excessive amounts of toxic pesticides for control of phytopathogens and insects to minimize crop losses. However, injudicious application of these pesticides causes environmental pollution, development of resistance in pathogens, disturbs ecological biodiversity and even causes public health hazards [5]. Hence, a sustainable and economical option is needed for controlling plant diseases and improving crop yield. Recently, biological control of pathogens and pests by the use of antagonistic microorganisms (biocontrol agents), their extracted bioactive metabolites or organic amendments has emerged as an environment friendly and cost effective technology for controlling/killing the pathogens and suppression of plant diseases [269, 270]. These biological methods of controlling phytopathogens will decrease the excessive use of agro-chemicals in crop fields and will reduce the pollution as well as other health related problems [271–273].

In a stride to assess the significance of organic amendments on biological control of phytopathogens, different amendments either singly or in combination with biocontrol agents were tested for suppression of the plant diseases. For instance, pyroligneous acid (PA) or wood vinegar (an acidic reddish-brown aqueous liquid), possessing antifungal, termiticidal and antimicrobial activities, has been produced through the pyrolysis of lignocellulosic raw materials [274, 275]. In another experiment, vermicompost application was found to reduce the number of parasitic nematodes and encouraged the fungivorous and bacterivorous nematodes in soil [131]. In another study, three organic soil amendments viz., cassava peel (*Manihot esculenta*, Cranz), sawdust (*Gmelina arborea*, Roxb) and leaves of Spanish cedar (*Cedrela odorata* L.) were evaluated on the stalk rot disease of maize (*Zea mays* L.) caused by *Fusarium verticillioides* [276]. Interestingly, maize plants subjected to cassava peels combined with *C. odorata* treatment showed significant reduction in disease index and disease severity due to favourable competition with pathogenic fungi *F. verticillioides*. All these soil treatments also showed significant growth promoting effects on plant height, leaves number, leaves surface area and stems girths as compared to control plants.

Ali et al. [229] carried out initial reductive soil disinfestation (RSD) and applications of *Bacillus* (RSD-SQR), *Trichoderma* (RSD-T37), and biochar (RSD-BC) reduced the incidence of wilt by 26.8, 37.5, and 32.5%, respectively in comparison to non-RSD treatments. Greater suppressiveness of *Fusarium* wilt and *F. oxysporum* was observed in treatments with RSD-SQR-T37-BC and RSD-T37. In similar studies, use of integrated pest management (IPM) approaches statistically decreased the *Fusarium* wilt severity of lettuce from 50 to 70% in comparison to untreated controls, in both fields under two consecutive years [277]. The following treatments were used: (i) compost enriched with *Trichoderma*, (ii) combination of "*Trichoderma gamsii* + *T. asperellum*, *Bacillus amyloliquefaciens* and potassium posphite", and (iii) combination of "*T. polysporum* + *T. arviridus*". An increased crop yield was also observed in comparison to untreated controls. High-throughput sequencing of bacterial and eukaryotic rRNA gene markers of rhizosphere microbiota was performed and it showed Proteobacteria, Firmicutes and Actinobacteria as predominant phyla for bacteria and Ascomycota for fungi.

Ikram et al. [278] evaluated the effect of biochar alone and in combination with different oil cakes (viz. castor, coconut, mustard, linseed, and sesame) on tomato (*Solanum lycopersicum* L.) plants cv. Pusa–Ruby. It was observed that biochar in combination with various oil cakes developed resistance to the root-knot nematode *Meloidogyne incognita* in tomato (*Solanum lycopersicum* L.) plants. Biochar amendment with oil cakes also enhanced plant growth, fresh weight to dry weight, total chlorophyll, carotenoid levels, nitrate reductase activity, and reduced the eggmass/root, the number of galls and nematode population in comparison to the untreated control plants. Biochar + castor oil cake showed a higher effect in the declining nematode population, followed by biochar + mustard cake, biochar + sesame cake, biochar + linseed cake, and biochar + coconut cake. When biochar was used alone, the effect was minimal. Among all treatments, the amendment of biochar and castor oil cake has the highest ability to manage the root-knot nematode infestation in tomatoes. In another study, Yan et al. [147] showed that biochar and biochar combined with *Bacillus* synthetic community (SynCom) alleviated replant disease in *Radix pseudostellariae* by modulating rhizosphere soil protistan and plant microbial communities. Field experiments showed that biochar alone and combined with *Bacillus* SynCom improved the physiological traits and main components of *Radix pseudostellariae*. The combined treatment significantly decreased the relative abundance of pathogenic *Ralstonia*, promoted the abundance of *Bacillus* in the root and *Paenibacillus* in the leaf, and increased the abundance of the predatory protists *Cercomonas* and *Paracercomonas* in the soil.

5.5 Remediation of pesticides and organic pollutants

For the management of pathogenic fungi, weeds, and insect pests in crop fields, a variety of pesticides, including fungicides, herbicides, and insecticides, are often utilized. Nevertheless, fewer than 1% of the insecticides used in field applications actually reach the pests [279]. The residual pesticides harm the environment and human health because of their toxicity and environmental pollution [5]. In addition, various organic contaminants in agricultural soils including detergents, dyes, lipids, phenols, oil and surfactants are continually being added in the soil through various processes. Thus, the occurrence of pesticides and organic pollutants in soil is a major environmental concern. Use of single or groups of bacteria, microbial-algae consortia, or organic amendments (compost or biochar) has been recommended as a sustainable approach to decontaminate or detoxify the pesticides and organic pollutants from the environment [46, 75]. Ling et al. [280] reported that amendment of soil with Biochar500 or Biochar600 (pyrolyzed at 500 °C and 600 °C, respectively) induces a shift in microbial community structure towards dominance of Gemmatimonadetes, Bacteroidia, Alphaproteobacteria and Gammaproteobacteria classes and promotes faster mineralization of SOC, thus offsetting C sequestration effects. The majority of soil derived CO₂ efflux in biochar amended soils originated from recalcitrant components that were mineralized by the persistent organic matter decomposers.

The application of biochar and wood vinegar together prevented glyphosate pesticides' hazardous leaking into the aquatic environment [281]. In another study, humic acids exhibited a potent capacity to sorb cationic and hydrophobic organic pollutants as it was shown that the functional groups of humic acid, such as quinones, may establish covalent connections with aromatic amines or similar chemical molecules [282]. According to certain reports, lignin-modifying enzymes (LMEs) such laccase and peroxidase can catalyse the oxidative coupling of 2,4-dichlorophenol to humic and fulvic acids [283]. These hazardous residues that have been covalently bonded are efficiently detoxified since they are not bioavailable. Thus, application of humic substances (HSs), LMEs or interactions between microorganisms' and HSs could offer a useful strategy for the biodegradation of organic pollutants and remediation of pesticides [127]. Recently, Ni et al. [284] reported that combined application of corn straw biochar (CB, 2%) and low-molecular-weight organic acids (LMWOAs, 10 mg kg⁻¹) dissipated 86.58% of napropamide (amide herbicide) after 1 month of incubation. The napropamide in the soil was significantly absorbed by CB throughout this process, which drastically decreased its bioavailability. The combined treatment boosted the relative abundances of microorganisms that take part in the biodegradation of napropamide, accelerating the breakdown of xenobiotics in soil.

Zhou et al. [224] investigated the influence of biochar on toxicity mitigation and microbial community succession in soil contaminated by polycyclic aromatic hydrocarbon (PAH)-phenanthrene. The biochar-amended treatments (BC) and biochar immobilized with the phenanthrene-degrading bacteria (MBC) showed significantly lower phenanthrene removal rate (PRR) than the no-biochar treatments i.e., no biochar or inoculation (CK) and inoculation of phenanthrene-degrading bacteria (MB). The plant dry weight and height of the maize plants treated with BC, MB, and MBC were significantly higher than those of the CK treatment, and the highest dry weight was observed in the MBC treatment (biochar immobilized with the bacteria). Biochar amendment significantly changed the bacterial community structure and strengthened the bacterial co-occurrence network. Furthermore, biochar amendment alleviated the toxicity or stress of PAH on soil microbes. Bouhia et al. [32] isolated fungal species, FUN 06 (*Galactomyces geotrichum*), from olive

mill wastewater sludge evaporation ponds and tested its suitability for bioremediation and transformation of a mixture of OMWS and rock phosphate (RP) into a high-value organic fertilizer. The fungal inoculation notably diminished OMWS phytotoxicity while enhancing its physicochemical parameters, nutrient concentrations and removal of toxic organic compounds by up to 90% in comparison to the untreated control, and enhanced plant growth. Thus, showed promise for effective bioremediation and sustainable reuse.

In a recent study, Lin et al. [285] reported that amendment of biochar (1.0%) caused a significant increase in rhizosphere pH, organic matter and humus, and achieved a satisfactorily higher DEHP removal efficiency (maximally 77.53%). Biochar addition also remarkably changed rhizosphere bacterial communities by enriching some potential DEHP degraders of *Nocardioides*, *Sphingomonas*, *Bradyrhizobium* and *Rhodanobacter*. The abundance of genes encoding key enzymes (hydrolase, esterase and cytochrome P450) and DEHP-degrading genes (*pht3*, *pht4*, *pht5*, *benC-xyI/Z* and *benD-xyI/L*) were increased after biochar amendment, leading to the change in DEHP degradation metabolism, primarily from benzoic acid pathway to protocatechuic acid pathway. These studies provided clues for the mechanisms of biochar-assisted DEHP degradation in organic contaminated farmland soils.

6 Plant growth stimulation and increases in crop yield

Amendments of soil with compost, vermicompost and biochar have been established as a effective method for increasing crop yields, while lowering the need for chemical fertilizers, leading to sustainable agriculture [37, 57, 63]. In addition, 10–40% photosynthetically fixed carbon is deposited within the rhizosphere primarily as root biomass, exudates, and microbial biomass as soil SOM [129, 239]. Moreover, combined application of organic amendments alongwith single or consortium of beneficial microbes has been found to enhance biomass and yield of different crops under field and greenhouse conditions [37, 182, 286] (Table 2). The synergistic effects of organic amendments with beneficial microbes increase the accessibility of nutrients including N, P, K, Zn and S, mitigate abiotic stresses and suppress plant diseases. Thus, supply of carbon and energy resources from decomposed plant residues in the form of SOM alongwith root exudates help in proliferation of beneficial microbial community. Subsequent expression of plant growth promoting activities of beneficial microbes leads to stimulation of plant growth and increases in crop yield, contributing to circular sustainable agriculture systems. But in some instances, some limitations have been reported, which restrict crop development in a variety of agricultural settings. For example, the survival, viability and functionality of inoculated microbes under field conditions are often different under laboratory and greenhouse environments due to competition with indigenous microbes and it adversely impact the growth of crop plants [287]. Therefore, organic compounds, cell protectants, and nanoparticles are frequently added to beneficial biofertilizer strains to boost their survival and effectiveness for improving crop yield [33].

The effectiveness of various organic amendments including compost, vermicompost and biochar on yield of different crop varied depending on soil management, SOM content, rate of application and frequency of chemical fertilizers in integrated nutrient management (INM) system as well as on the addition of other soil amendments. For instance, addition of municipal waste compost at a dose of 80 tonnes ha⁻¹ annually for 5 years, enhanced the wheat yield by 246% in comparison to untreated control in a field study [288]. In similar studies, Erhart et al. [289] reported 10% increase in the yields of cereals and potato crop by application of biowaste compost (23 tonnes ha⁻¹ year⁻¹) compared to unfertilized control treatment. Using INM approach, Diacono and Montemurro [290] showed that application of organic amendments along with mineral fertilizers increased the plant growth and crop yield compared to application of soil amendment alone.

Recently, the development of ecological circular agriculture system has highly encouraged the use of biogas slurry as an important kind of fertilizer owing to its high nutrient components, environmentally friendly and lower costs than chemical fertilizers. Biogas slurry can be added directly to the crops or it may be used along with other organic amendments and synthetic stimulants. For example, Yu et al. [118] observed that amendment of biogas slurry in soil improved the soil texture and enhanced the crop yield and fruit qualities of tomato. Similar studies carried out by Pathak et al. [291] showed that seeds soaked in biogas slurry improved germination rates and increased seedling growth than untreated seeds. Zhao et al. [292] sprayed biogas slurry on *Capsicum* plants in the field and it increased crop yields and prevented the diseases. Addition of animal urine along with biogas slurry was found to increase its N content, resulting in increased C:N ratio for easy nutrient accessibility to plants and soil microflora. On the other hand, long-term application of untreated biogas slurry in crop lands has been found to adversely affect the crop yield, resulting in losses of crop yield and quality [293, 294]. The continuous application of biogas slurry has also been reported to increase the accumulation of heavy metals or genes for antibiotic resistance in soil [295, 296]. It may also cause leaching or movement of extra nutrients into

Table 2 Effect of different organic amendments (OAs) on growth of various crop plants and their mechanisms of action

Organic amendment	Origin of OA; dose	Application with	Crop; growth conditions	Mechanism of action	References
Compost	Rice straw	Sewage sludge	<i>Hordeum vulgare</i> ; Greenhouse	Improved soil quality and fertility	Roca-Pérez et al. [309]
	Rapeseed residues	–	Rice; Greenhouse	Reducing the bioavailability of Cd and Pb; improved soil quality and fertility	Ok et al. [310]
Vermi- compost	Green waste; 5% (w/w)	Arbuscular mycorrhizal fungi	Date palm (<i>Phoenix dactylifera</i> L.)	Increased photosynthetic efficiency, stomatal conductance and leaf water connections	Ait-El-Mokhtar et al. [311]
	–	<i>Azotobacter chroococcum</i> and <i>Piriformospora Indica</i>	Rice; Greenhouse	Improved plant nutrition especially available P	Prajapati et al. [312]
	Cow dung (3 t ha ^{−1})	–	Rice; Field	Increased soil microbial biomass and enzyme activity	Tejada and González [313]
	Rice and maize straw	Biochar	Wheat; Field	Improved water retention capacity, chlorophyll content, leaf K ⁺ content and reduced oxidative stress	Hafez et al. [314]

Table 2 (continued)

Organic amendment	Origin of OA; dose	Application with	Crop; growth conditions	Mechanism of action	References
Biochar	Rice straw; 2.5% (w/w)	–	Ryegrass; Field	Increased soil available P and K	Zhang et al. [315]
	Rice husks; 21 g kg ⁻¹	<i>Bacillus pumilus</i>	Rice; Greenhouse	Increased soil total C, C/N ratio, exchangeable K ⁺ , chlorophyll content and nutrient uptake	Win et al. [316]
	Rice straw; 15 t ha ⁻¹	Jasmonic acid	Faba bean; Greenhouse and pots	Reduced oxidative damage to leaf tissues and maintenance of the integrity of cell membranes; salt stress mitigation	El Nahhas et al. [317]
	Rice hull; 1 t ha ⁻¹	PGPR <i>Bacillus</i> spp.	<i>Radix pseudostellariae</i> ; Field	Changed rhizosphere soil metabolites and stimulated proliferation of beneficial microorganisms; alleviation of biotic stress	Wu et al. [318]
	Wheat straw; 20–40 t ha ⁻¹	–	Maize; Field	Increased N uptake	Zhang et al. [319]
	Wheat straw; 1 t ha ⁻¹	–	Wheat and rice; Field	Improved soil aggregation and fungal community structure	Bai et al. [320]
	Wheat straw; 2% (w/w)	–	Tomato; Greenhouse	Increased N and P uptake; growth suppression of <i>Ralstonia solanacearum</i>	Tian et al. [321]
	Wheat straw; 20 t ha ⁻¹	–	Rice; Field	Increased N and P uptake	Liu et al. [322]
	Maize straw; 15.75–31.5 t ha ⁻¹	–	Soybean and maize; Field	Improved soil aggregation and increased soil organic carbon content	Jin et al. [323]
	Maize straw; 5% (v/v)	–	Maize; Greenhouse	Mitigation of salt and water stress; Improved antioxidant defense machinery in plant and enhanced nutrient uptake	Ndiate et al. [324]
	Maize straw; 2% (w/v)	–	Maize; Greenhouse	Increased N uptake	Xia et al. [325]
	Barley straw; 10 t ha ⁻¹	Inorganic fertilizer	Chinese cabbage; Field	Increased N, P and K uptake	Kang et al. [326]
	Barley straw; 20 t ha ⁻¹	Inorganic fertilizer	Rice; Field	Improved soil physical properties and increased soil chemical contents	Kang et al. [327]

Table 2 (continued)

Organic amendment	Origin of OA; dose	Application with	Crop; growth conditions	Mechanism of action	References
Biogas slurry	–	Exogenous Ca, Mg and P nutrients	Spathiphyllum	Enhanced N assimilation	Liang et al. [328]
Other amendments	Cow dung	–	Tomato; Greenhouse	Increased fruit yield; Reduced nematode attack	Jothi et al. [329]
	–	–	Maize; Field	Improved biomass and protein content	Islam et al. [330]
	<i>Rhizophagus irregularis</i>	–	<i>Phragmites australis</i> ; Field	Phyto–rhizoremediation	Wu et al. [331]
	AM fungi i.e., <i>Rhizophagus fasciculatus</i> , <i>R. intraradices</i> , <i>Funneliformis mosseae</i> , <i>Glomus aggregatum</i>	–	<i>Zea mays</i>	Phytoextraction	Singh et al. [332]
	Compost + AMF consortium	PGPR	Tomato	Improved growth and yield, and drought tolerance	Tahiri et al. [333]
	Rose distillation waste	Mycorrhizae and PGPR	Rose seedlings	Increased leaf area, stem height growth and bud break rate	El Malahi et al. [308]

groundwater or surface water. Therefore, pre-treatment of biogas slurry has been suggested as safe strategy before its application in farmlands.

Schmidt et al. [297] reported that amendments to a vineyard soil with biochar alone (8 t ha^{-1}), aerobic compost (55 t ha^{-1}) and biochar-compost ($8 \text{ t ha}^{-1} + 55 \text{ t ha}^{-1}$) induced small and non-significant effects on fruit quality, health of plants, nutrient absorption, and development of plants over the three years. Vine growing in less fertile, alkaline and temperate soil conditions did not immediately benefit economically from applying more biochar to the top soil. Ding et al. [41] demonstrated that typically biochar has a large surface area with many functional groups, a high cation exchange capacity, and great stability. Therefore, biochar application to soil was suggested to play critical role in enhancing the fertilizer use efficiency, nutrient availability and exchange of nutrients. Thus, biochar application increases the crop yield. However, before it is feasible to precisely evaluate the sustainability of this strategy to trap carbon and restore soil function, further research on the long-term effects of adding biochar on soil characteristics is still needed.

The co-application of wood vinegar and biochar was found to enhance the fertility of soil and also promoted plant development by enhancing seed sprouting and seedling development [298]. Yu et al. [299] presented an overview about problem soils (that inhibit or prevent plant growth) and application of biochars as useful soil amendment materials. The interactions of biochar with soil particles, soil microflora and roots of plants were discussed in relation to remediation of pollutants from contaminated soil and improvement in plant growth. Authors suggested that adding biochar to contaminated soils is a practical strategy to improve their quality and boost crop productivity. Bouhia et al. [228] showed that addition of biochar to compost increased the levels of macro- and micro-nutrients compared to those under single application of biochar. The soil fertility improved significantly with regard to available P and K, N, and micronutrients with the addition of biochar singly or in biochar-compost combination in *Zea mays* grown in pots. Overall, single application of compost gave the best results in terms of plant growth and soil fertility improvement. In another study, Ebrahimi et al. [300] reported that soil amendment with vermicompost and pistachio biochar and 100% plant water requirement (PWR) showed the best plant growth of eggplant (*Solanum melongena* L.). Application of both pistachio biochar and vermicompost at 100% PWR resulted in the highest leaf concentrations of N, P, K, Fe and Mn. The total plant yield was found highest under combined application of vermicompost and pistachio biochar at 100% PWR. Salvador et al. [187] reported that organic compost containing *Bacillus subtilis* resulted in a 50% increase in root dry weight for soybean plants in comparison to soils without any amendment, whereas plants grown in *Rhizophagus irregularis* soil showed higher nodule biomass (~30%), symbiotic efficiency (~50%), chlorophyll content (~20%) and nitrogen accumulation (~45%).

In a stride to establish the role of organic amendments on yield attributes and to reduce drought stress-induced damages in wheat (*Triticum aestivum*), Anee et al. [81] added different OAs consisting of compost, vermicompost and poultry manure @ 0.09 kg m^{-2} soil, biochar @ 2.5% w/w soil and chitosan @ 1% w/w soil. Addition of vermicompost enhanced soil plant analysis and development (SPAD) value by 18%, number of spikelets spike⁻¹ by 20%, number of grains spike⁻¹ by 17%, whereas poultry manure amendment increased plant height by 16% under drought condition compared to control plant. Vermicompost also showed the capability to reduce the level of lipid peroxidation and raised the proline content by drought condition. Drought stress-induced increment of catalase, ascorbate peroxidase and glutathione reductase activities were also efficiently modulated by the organic amendment application. Bornø et al. [301] used wood biochar or straw treatments to maize seedlings that were subjected to various irrigation schedules. In the early drought and recovery phases, biochar additions had a beneficial impact on leaf water potential and photosynthetic metrics, but had no impact on plant biomass. Additionally, biochar impacted the pH and ammonium content of the soil, and also boosted the C-substrate utilization in the biochar treatments, irrespective of watering. Drought particularly had an additive effect on the level of succinic acid in the wood biochar treatment, whereas biochar specifically raised the content of organic acids.

In similar studies, Ghorbani et al. [302] examined wheat's ability to utilize nitrogen in connection to azocompost (A), biochar (B) combined with urea (U), and leftovers from legumes (L). The treatments included: "Control (no biochar, no fertilizer), U (46 kg ha^{-1}), A ($5 \text{ tonnes (t) ha}^{-1}$), L (5 t ha^{-1}), B (10 t ha^{-1}), UB, AB, and LB". With UB and UL treatments, the grain production of wheat was increased by 337% and 312%, respectively. The greatest grain nitrogen yield was achieved by the LB, rising 8.8 times above the control. In comparison to the control, legumes had the highest efficiency in nitrogen use, increasing by 149%. With a value of 91 and 70%, respectively, the use of LB resulted in the best N-harvest index and N-recovery efficiency. Khucharoenphaisan and Sinma [101] observed that addition of rice straw compost (at a dose of 6 tonnes ra^{-1}) showed better growth promotion ability of asparagus (*Asparagus officinalis*) than peanut straw compost and sugarcane bagasse compost. Application of rice straw compost and chemical fertilizer (30 kg ra^{-1}) gave the cumulative number of shoots i.e., 17.2 and 14.0 shoots plant⁻¹, respectively. Sharma et al. [303] produced high-quality stabilized compost from biodegradation of solid waste using decentralized in-vessel composting and its application improved the growth yield of *Brassica juncea* (mustard), *Spinacia oleracea* (spinach) and *Raphanus sativus* (radish).

The combined application of plant feedstock biochar and vinegar improved the yield of tomato (*Solanum lycopersicum* L.) plants [304]. In contrast, vinegar application could potentially limit the yield and fruit sugar content of tomato plants treated with animal feedstock biochar. The sugar content was higher for the chicken manure and animal litter biochar, which was attributed to better pH conditions. Rajput et al. [254] reported that individual application of biochar (2.5% of total soil) and/or bacterization with metal-tolerant bacteria (10^{10} CFU kg⁻¹) on barley (*Hordeum sativum* L.) reduced the uptake of zinc in roots by 44% and 20%, and those in the plant's above-ground portions by 39% and 13%, respectively as compared to contaminated soil treatments with ZnO nanoparticles (ZnO NPs; 2000 mg kg⁻¹ concentration). Separately and in conjunction with bacteria, biochar improved plant height by 53% and 40%, and root length by 48% and 85%, respectively in comparison to the contaminated control. A decrease by 52% and 40% was observed in the root length and plant height in soils spiked with ZnO nanoparticles compared to clean soil treatments. Treatment of contaminated soil with metal-tolerant bacteria decreased the zinc compound mobility by 7%, whereas addition of biochar lowered mobility of zinc mobile components by 33%. By applying both biochar and bacteria together, the mobility of zinc mobile compounds was reduced in soil.

Nawaz et al. [305] showed that drought-mediated decrease in photosynthetic pigments and gas exchange characteristics was markedly lower in biochar (BC) and PGPR-treated soybean plants in pots under semi-controlled conditions in comparison with control. Plants treated with BC + *Paraburkholderia phytofirmans* showed maximum increase in photosynthetic (12–30%) and the enzymatic antioxidant activities (5–20%), whereas no further significant increase was observed in plants treated with combined BC and *Bacillus* sp. under water deficit conditions (25–30% water holding capacity). Similarly, the combined BC and *P. phytofirmans* application considerably increased the grain yield (14%) under drought stress conditions. In another study, Behzadi et al. [306] investigated the impact of vermicompost and foliar application of zinc sulfate under conditions of reduced irrigation on the physiological properties of borage (*Borago officinalis* L.). An increase in vermicompost fertilizer application led to an elevation in relative water content (RWC), with the highest RWC (79.2%) achieved when 10 ton ha⁻¹ of vermicompost was applied. Vermicompost supplementation improved chlorophyll a + b content during the seed-filling stage under conditions of irrigation cut-off. Furthermore, the findings revealed that applying 2 and 4 mg l⁻¹ zinc sulfate reduced malondialdehyde content by 5% and 9%, respectively. The catalase, peroxidase, and superoxide dismutase activities demonstrated an increased response to stress mitigation treatments. However, their activities decreased as vermicompost and zinc sulfate levels increased. The study demonstrated that the highest biomass was obtained when 10 ton ha⁻¹ of vermicompost and 2 mg l⁻¹ of zinc sulfate were applied. Using vermicompost and zinc sulfate, particularly during the seed-filling phase, alleviated the adverse effects of drought stress.

Significant increases in the soil pH, organic matter and total N content were reported by higher application rate of biochar (at the dose of 2.5 and 5% biochar level) in soil cultivated with rice [307]. Biochar was found to affect soil microbial community structure/functions, and its higher application rate decreased the soil bacterial diversity but elevated the fungal diversity. However, dominant species taxa were not altered by application of different concentrations of biochar. Thus, biochar application was found to increase rice crop yields by regulating soil microbial taxa composition. However, more research has to be put into figuring out how the different levels of biochar application differentially affect physiological functions of the soil microbes, soil nutrient indexes, crop yields and ecosystem's health. In another recently conducted experiment, biochar derived from tomato straw (TSB), sweet pepper straw (SPSB), and eggplant straw (ESB) was analyzed for nutrient content, cation exchange capacity and adsorption rates [252]. SPSB showed significantly higher levels of N, P, and K, along with superior adsorption capacity compared to TSB and ESB. Treatment of cucumber with 5% (w/w) SPSB concentration increased plant height by 13.01%, stem thickness by 20.79%, leaf area by 50.26%, and dry weight by 58.56% in comparison to the untreated control under greenhouse conditions.

Another recent study highlighted the potential of floral waste as sustainable alternative to traditional synthetic chemicals for use in organic agriculture. For example, El Malahi et al. [308] demonstrated that application of rose distillation waste on seedlings of *Rosa damascena* showed a significant increase in terms of leaf area and a maximum apical internode growth of 1.5 mm was observed in seedlings. The incorporation of biostimulant (containing both mycorrhizae and plant growth promoting rhizobacteria) in the rose distillation waste substrate resulted in significant stem height growth of 11.3 cm and a high bud break rate of 73.3%. In similar studies, inoculation of earthworms *Pontoscolex corethrurus* (Rhindrilidae) significantly enhanced the fertilizer performance of different types of organic amendments (OAs) [135]. Overall, OAs with low fertilizer performance showed a more pronounced response to earthworms. Additionally, earthworms improved the performance of high-quality OAs likely by enhancing the availability of P and Mg, suggesting synergistic effects of earthworms and OAs of various qualities on rice plant growth in nutrient-poor Ferralsols of Madagascar. The rice grain yield increased with application of biochar at application rates of 1.5% (T1.5), T3.0, and T4.5 as compared to control untreated (T0) by 116.38, 141.24 and 145.20%, respectively in a saline-alkali paddy field [212].

7 Future perspectives and scope

Significant increases in the crop yields and quality of food production have been achieved by application of modern agricultural technologies. Some of these technologies utilize chemical fertilizers, pesticides and plant growth regulators for securing sufficient crop yields to feed over seven billions of human beings on the earth. However, extensive and injudicious uses of these agrochemicals have negative impacts on the environment, microbial diversity and human health. Therefore, there is urgent need for sustainable agriculture, which relies on a better understanding of organic waste management as well as beneficial interactions between plants and soil-inhabiting microorganisms.

Organic wastes can be processed and utilized in different forms and for various purposes in agriculture. They are mainly utilized for improving soil's physical and chemical properties, and for nutrient replenishment. In the last few decades, soil contamination with organic and inorganic wastes, pesticides and HMs is also gaining much attention of researchers. Multiple technologies have been devised to manage agricultural waste and to overcome soil pollution [25, 31, 334]. But, all these technologies have some demerits when used individually. Therefore, there is a need to develop integrated techniques or to modify existing methods to get maximum benefits for improving soil fertility and crop production. The modifications or improvements must concentrate to develop sustainable methods, not damaging the environment in the long run, easy to use, ethical, economical and affordable.

Most of the organic waste management practices and soil pollution mitigation methods employ different microorganisms at certain stage of agri-technology [28, 110, 113]. The selection of particular microbial strain is the crucial step of any bioconversion technology. Although, microorganisms exhibit remarkable evolutionary capabilities but their natural evolution is relatively slow. In such scenarios, bioengineering of beneficial microorganisms may be helpful to enhance the bioconversion capacity of pollutants [335]. Recent developments in Microbiology, Molecular biology and Biotechnology are providing the driving force towards engineering of better microbes for bioremediation of polluted soil and to obtain improved microorganisms with desired characteristics. Mishra et al. [336] discussed the profiling of microorganisms present in polluted sites using various techniques including culture-based and omics-based approaches. The use of microbial electrochemical technologies was also reviewed for remediation of pollutants. In addition, the effects of various treatment technologies for remediation of pollutants were evaluated and future outlook and challenges were discussed.

The use of plant biostimulants and microbiome engineering has recently emerged as environment-friendly technology for improving nutrient use efficiency and mitigating biotic and abiotic stress resulting in plant growth promotion and increases in crop yield [159, 178]. Recently, genetically modified organisms (GMOs) or bioengineered microbes have been developed for faster degradation and removal of soil contaminants such as pesticides, hydrocarbons, and HMs to restore soil health [335]. Applications of GMOs were found to improve the bioremediation process by altering metabolic pathways that results in hyper-secretions of different biomolecules. In order to make the soil productive and healthy, contaminated or polluted soils may be treated with GMOs, PGPR, AMF and other soil additives for effective restoration of soil health [147, 333]. The resource efficiency may further be enhanced by appropriate waste management and valorization processes of agricultural waste biomass into value-added products for execution of sustainable circular bio-economy [337, 338]. Moreover, the synergistic effects of organic amendments with beneficial microbes may become a preferred management option to provide valuable insights for fertilization strategies in nutrient-poor ecosystems, benefiting both plant growth and crop yield components.

8 Conclusions

Production of organic agricultural wastes is unavoidable due to increasing food demand for exponentially growing human population worldwide. But, the accumulation of unprocessed organic waste on soil is causing environmental pollution. Different strategies and technologies used for organic waste management include composting, anaerobic digestion, vermicomposting, biochar production, organic phytochemicals, and bioremediation of pollutants. Application of these processed organic wastes in the farm soil increased nutrient availability, stimulated the population of beneficial microbes, caused remediation of heavy metals, inhibited the growth of phytopathogens, and detoxified or degraded the pesticides and organic pollutants. Furthermore, fertilization of soil or crops with these processed organic materials improved soil fertility and health; and stimulated plant growth resulting in increases in crop yields.

Thus, appropriate management of organic wastes by using recently developed technologies will not only reduce the application of chemical fertilizers in fields, but will also help the soil to restore its natural physical, chemical and biological properties. Further research on the current and future valorization routes of organic wastes, from detoxification of pollutants and pesticides to the development of promising applications in agriculture, energy, and the environment will have significant implications for low-income countries. However, inconsistent results obtained by the application of biochar, compost and vermicompost in polluted or degraded soils for improving crop productivity and limited understanding of complex plant-microbiome-environmental interactions in soil are still few limitations. More field experiments could help in upscaling the utilization of organic amendment technologies for better understanding of plant-microbe interactions in the field conditions and for harnessing better plant growth and crop yields in sustainable agriculture.

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Declarations

Ethics approval and consent to participate The work submitted does not include any experiment related to animals.

Informed consent No conflicts, informed consent, or human or animal rights are applicable to this study.

Competing interests The authors declare no competing interests.

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